A PRELIMENERY REPORT ON

SMART FARMING USING MACHINE LEARNING AND DATA ANALYTICS

SUBMITTED TO THE VISHWAKARMA INSTITUTE OF INFORMATION TECHNOLOGY, PUNE

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BACHELOR OF TECHNOLOGY (COMPUTER ENGINEERING)

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CERTIFICATE

This is to certify that the project report entitles

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ABSTRACT

'Smart farming using machine learning and data analytics' is a POC intended for betterment of farmers and agriculture in India. Mentioned in the report are the exact goals to be fulfilled by this project – namely, recommendation of the top n crops based on soil properties and environment factors. These take into consideration all the nutrients and micronutrients (around 12 of them) with reference to the soil health card. From these top n crops recommended, the farmer will also be able to know the economically beneficial crop where the model implements price prediction module for the crops.

Also a year round plan will be provided to farmer to maximize his cultivation and earnings through CSM. Besides generic information regarding government schemes or soil testing labs will be provided. The researcher along with these functionalities can access certain datasets that he wants to explore. The researcher can also study the impact of various algorithms on varies data. An add on 'what if' scenario will help not only the farmer but also the researcher to play around with various factors and see the variations in results and proceed with decision making accordingly.

The literature survey done for the same is mentioned in further parts of the report. Around 7 IEEE papers have been referred to for the study and framing of the problem statement. Loopholes amongst the papers or future work mentioned in them provisioned the project with gap identification and thus trying to resolve the same. We move to the software requirements part where the things necessary to develop the project were understood, studied and tried to accomplish. The system design part provides an end to end idea of what exactly the project would look like. It also mentions the overall architecture of the project and other aspects of software engineering.

Project implementation states the overall algorithms used, the equations formed for the project that will yield better results. Also the tools and technologies used or the platforms and environments worked in have been specified. For a project to be actually stable and accepted in real world scenario it should bypass many test cases. The testing part and for what aspects the project has been tested is stated. The report concludes with the results including a web app portal for farmer and researcher to use which internally is trying to maximize the accuracy of prediction. A lot of things still remain that have to be provisioned through this web app. This

gives us the potential for future work and future scope that has been mentioned. Solving a real world problem, this web app can be used for other applications by simply altering the datasets, such applications have been specified

Thus an effort in solving a real world problem with a lot of uncertainty in factors affecting it. Also exploration of various front end technologies, back end ones and algorithms done to solve the same.

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CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

'Smart farming using machine learning and data analytics' is a POC intended for betterment of farmers and agriculture in India. Mentioned in the report are the exact goals to be fulfilled by this project – namely, recommendation of the top n crops based on soil properties and environment factors. These take into consideration all the nutrients and micronutrients (around 12 of them) with reference to the soil health card. From these top n crops recommended, the farmer will also be able to know the economically beneficial crop where the model implements price prediction module for the crops.

Also a year round plan will be provided to farmer to maximize his cultivation and earnings through CSM. Besides generic information regarding government schemes or soil testing labs will be provided. The researcher along with these functionalities can access certain datasets that he wants to explore. The researcher can also study the impact of various algorithms on varies data. An add on 'what if' scenario will help not only the farmer but also the researcher to play around with various factors and see the variations in results and proceed with decision making accordingly.

The literature survey done for the same is mentioned in further parts of the report. Around 7 IEEE papers have been referred to for the study and framing of the problem statement. Loopholes amongst the papers or future work mentioned in them provisioned the project with gap identification and thus trying to resolve the same. We move to the software requirements part where the things necessary to develop the project were understood, studied and tried to accomplish. The system design part provides an end to end idea of what exactly the project would look like. It also mentions the overall architecture of the project and other aspects of software engineering.

Project implementation states the overall algorithms used, the equations formed for the project that will yield better results. Also the tools and technologies used or the platforms and environments worked in have been specified. For a project to be actually stable and accepted in real world scenario it should bypass many test cases. The testing part and for what aspects the project has been tested is stated. The report concludes with the results including a web app portal for farmer and researcher to use which internally is trying to maximize the accuracy of prediction. A lot of things still remain that have to be provisioned through this web app. This

gives us the potential for future work and future scope that has been mentioned. Solving a real world problem, this web app can be used for other applications by simply altering the datasets, such applications have been specified

Thus an effort in solving a real world problem with a lot of uncertainty in factors affecting it. Also exploration of various front end technologies, back end ones and algorithms done to solve the same.

1.2 MOTIVATION

India being an agricultural country, its economy is mainly and predominantly dependent on agriculture yield growth and also allied agro-industry products. In India, agriculture is essentially influenced by rainwater which is very unpredictable. Growth in agriculture influentially depends on various parameters of soil, namely Nitrogen, Phosphorus, Potassium and various micro nutrients. It also depends on Crop rotation, Soil moisture, soil type, pH, surface temperature, etc and weather aspects like temperature.

India now is rapidly progressing towards technical development. Thus, technology will convince to be beneficial to agriculture which can increase crop productivity leading to better yields to the farmer. The proposed project provides an answer for Smart Agriculture by monitoring the agricultural field which may assist the farmers in increasing productivity to an excellent extent.

This work presents a system, in a form of a website, which uses Machine Learning techniques in order to predict the most profitable crop based on the current weather, soil conditions, and predicted future market price. The proposed system tries to integrate the information obtained from various repositories, weather departments and by applying machine learning algorithms a prediction of best suited crops consistent with current environmental conditions is made.

This provides a farmer with varied options of crops that can be cultivated. Thus, the project generates a system by integrating data from various data sources, and focuses on data analytics, prediction analysis which may improve crop yield productivity and thus increase the profit margins of farmers helping them over a longer run.

1.3 PROBLEM DEFINITION AND OBJECTIVES

Problem Definition:

To apply Data mining and Machine Learning Techniques to solve the problems faced in agriculture domain which will:

- Provide the farmers with economically most beneficial crops.
- Provide farmer with options of top n crops suited for his land.
- Provide farmer a year round plan to maximize profit.
- Allow farmer and researcher to play along with various inputs, view its impact and proceed with decision making accordingly.
- Provide the information about various Government Schemes and Soil Testing Laboratories.
- Provide and allow the researchers to analyse the data.

Objectives:

- To apply Data mining and Machine Learning Techniques to solve the problems faced in agriculture domain.
- To compare the outputs of various algorithms.
- To increase the Return on Investment (ROI) for farmers.
- To study latest Algorithms
- To provide open source data to researchers

1.4 PROJECT SCOPE AND LIMITATIONS

Scope:

- The granularity of average rainfall will be restricted to minimum district level.
- Recommendation of crops using various soil parameters such as Nitrogen, Phosphorus, Potassium, micro nutrients, Crop rotation, Soil moisture, pH.
- Other parameters – Surface temperature, soil type and Weather aspects like temperature

Not in Scope:

- The granularity will not be below district level for average rainfall prediction.
- The cultivation assistance will not be given by the system.

- Only basic crops will be considered. (Horticulture will not be considered).
- IOT based sensors will not be used to collect data. Work will be done on data collected directly**.**

Limitations:

- The farmer must have the nutrients data.
- This implies he must have a soil health card or should be aware of his land properties.

1.5 METHODOLOGIES OF PROBLEM SOLVING

- Recognizing the need –
  Due to the unpredictable environmental factors or price patterns, farmers are unable to decide which crop to cultivate so as to maximize profit. Even if the crop suits his land a lot of it goes waste if there is no market demand. Thus a model was needed to guide the farmer considering his land and environment and price factors.
- Defining the objectives, problem, constraints –
  Thus as mentioned in 1.3 and 1.4 are the required points.
- Collecting information and data –
  Data required in the project was to train the module for prediction. The data is created by the project members from knowledge set – Various textbooks,
  Agriculture College Pune's App, Soil Health cards study, etc. Price data was taken from government sites.
- Generating alternative solutions –
  From user's point of view, he has been given the flexibility to choose amongst the various solutions provided by app – single or alternative crop, or a year round plan. From developing point more than 4-5 algorithms have been explored, considered, implemented to provide alternative solutions.
- Evaluating the consequence of different solutions –
  User might choose a solution as per his necessity. Algorithms were evaluated for accuracy and mean errors to improve accuracy and reduce errors.
- Deciding and specifying final best solution –

Farmer should go for economically beneficial or year round plan. Algorithm with highest accuracy used for base model. Researcher can refer to data he needs.

CHAPTER 2: LITERATURE SURVEY

2.1 BACKGROUND AND DOMAIN

India being an agro-based country, its economy mainly depends on agriculture produce growth and related agro-industry products. Indian agriculture largely depends on the rainfall which is highly unpredictable. Farming yield is also largely dependent on diverse and multiple soil parameters, like Nitrogen, Phosphorus, Potassium, Crop cycle, Soil moisture, soil pH, temperature and climatic aspects like temperature, rainfall, etc. Nowadays India is very rapidly progressing towards technical innovation and development. Thus, technology will be beneficial as well as blessing to agriculture which will enhance crop productivity which will result in better yields to the farmer.

The proposed project provides an innovative solution for Smart Agriculture by monitoring the soil nutrients and climatic conditions which can assist farmers in increasing productivity to a great extent. This project presents a system, in a form of a website, which uses Machine Learning techniques in order to predict three most suitable crops based on soil parameters and climatic conditions and also most profitable crop based on the predicted future market price. This system integrates the data collected from repository, Weather department and by applying machine learning algorithm a prediction of most suitable crops according to current environmental conditions is made. This will provide farmers with variety of options of crops that can be grown with enhanced yield. Thus, the project develops a system by integrating data from various sources, data analytics, prediction analysis which will improve crop yield productivity and ultimately increase the profit margins of farmers helping them over a longer run.

Agriculture

Different types of soils have various mineral contents and every crop require multiple mineral components for its better growth. Each and every soil has some specific characteristics and is suitable to grow only a particular number of crops. Therefore an agronomist should be known about the type of soil he has so that he could cultivate better crops. Another important issue that is famers have to face is the uncertainty in the market demand and prices from the time of sow to the time of harvest.

Machine Learning

Machine Learning is a section of Computer Science where new developments evolve at recent times, and also helps in automating the evaluation and processing done by the mankind, thus reducing the burden on the manual human power. It is a type of Artificial Intelligence that provides devices with the ability to learn without being separately programmed. It gives focus on the development of computer programs that can change when exposed to new data. Finding out the suitable crops based on the soil's appearance becomes tedious for novice farmers. There also exists a need to prevent the agricultural decay.

Data Analytics

It is a method of analysing, cleaning, transforming & creating models of data with the aim of finding meaningful and useful information, concluding that information and supporting decision making.

Data analytics projects and initiatives support businesses increase revenues, help in improving operational and functional efficiency, responds quickly to emerging market trends and gain a competitive edge over rivals, optimizes marketing campaigns and customer service efforts.

2.2 COMPARISON RESEARCH

1. Predictive Analysis to Improve Crop Yield using a Neural Network Model [IEEE 2018]

The architecture proposed in this paper provides a computational dimension to enhance knowledge about the yield before the crop sowing period. It provides unique model which performs joint prediction of both rainfall and soil features on the yield, which is also termed as hybrid model. The model reads historic data and learns through computational processes like ARIMA, Exponential Sequencing and Recurrent Neural Networks. It accepts soil features, fertilizer concentration and sowing period from the end user and provides expected yield in tonnes per hectare as output for given parameters. The approach not only promises

precise levels of expected yield but also helps determines seasonality trends in the yield of the same crop over different years in the same season. The proposed model is trained only for historic district-wise rainfall data and soil data of Karnataka state and predicts the yield of rice for same.

2. Crop Selection Method to Maximize Crop Yield Rate using Machine Learning Technique [IEEE 2015]

An agro-based country depends on agriculture for their economic growth. When the population of a country increases dependency on agriculture also increases and the subsequent economic growth of the country is affected. In this situation, crop yield rate plays a significant role in economic growth of the country. In addition to the biological approaches (eg. Seed quality of crop, crop hybridization) and some chemical approaches (eg. Use of fertilizer, urea, potash) approaches a crop sequencing technique is required to improve net yield rate of crop over season.

This paper proposed a new and unique method named Crop Selection Method (CSM) to achieve net yield rate of crops over season. This technique classifies the crops in 4 categories as Seasonal crops (eg. Wheat, paddy), Whole year crops (eg. Vegetables, paddy, toor), Short time plantation crops(eg. Potato, vegetables, rese), and long-time plantation crops (eg. Sugarcane).Algorithm provides unique combination of these crops that can be selected in a sequence based on yield rate per day as output. Method may improve net yield rate of crops using limited land resources and also increases re-usability of the land.

3. Crop Yield Prediction and Efficient use of Fertilizers [IEEE 2019]

The system provides a model precise in predicting the crop yield and also recommendations about the required fertilizers ratio based on soil parameters and atmospheric conditions to maximize the crop yield and farmers revenue. Among the factors for better crop yield, use of fertilizers is also an important factor as it directly affects the fertility of soil. Early detection of problems and providing their solutions will help in better crop yield. This paper predicts the crop yield based on two algorithms (Random Forest Algorithm and Back Propagation Algorithm) and compares the result of two algorithms so that the algorithms with more accuracy and less errors can be used. Also, the system provides the end user with the proper fertilizer ratio to be used to get the desired crop yield.

4. Decision Making Support System for Prediction of Prices in Agricultural Commodity [IEEE 2019]

Agriculture commodity prices have a variable nature impacting the profit negatively. Agriculture price prediction has always been a challenging problem for farmers. Farmers are unable to get the desired price for their crops due to which suicide rate is increasing year by year. Different social and economic factors are responsible for fluctuation of prices of agriculture commodity. This paper provides a decision making support model for prediction of prices in agriculture commodity which will help in reducing agribusiness risks. This paper uses soft computing techniques for predicting prices of various crops. Using data mining algorithms paper provides average price prediction for agriculture commodities helping farmers in decision making and ultimately profit making. Graphical comparison of prices of agriculture commodities is provided. The decision model takes previous sell price from farmer and predicts the future price.

5. Prediction of Future Market Price for Agricultural Commodities [International Journal of System and Software Engineering 2015]

Agricultural items have seasonal prices. These prices are spread over the year. If these prices are predicted in advance then it will be promising to the farmer on Return on Investment (ROI). The prediction of future market price of price may be valuable for budgetary planning systems for different government agencies. It will help in growth of economy of country like India where main source of income for larger population in agriculture. This paper provides a system which alerts the farmers about the crop he/she can sow such that by the time of harvesting he/she gets better return on investment. The market price of the crop from the time of sowing and harvesting can be uncertain. This paper makes use of neural networks (Neuroph framework) and back propagation algorithm for prediction of future price of the crops. The suggested model has the prediction engine which is the heart of the entire model. The paper provides the results in only graphical format and predicts the prices of only five crops (wheat, bajra, rice, paddy and jowar)

6. CROP PREDICTION USING PREDICTIVE ANALYTICS [IEEE 2017]

Theme:

In this paper, the author mainly focuses on aggregating various papers and further utilizing them to develop a model to predict suitable crops, crop yield and recommending fertilizers (future scope).

Model:

NPK sensors will be utilized to sense the nutrients data from soil. This data will be used to predict the best suitable crop based on soil nutrients. Further Agro algorithm will be used to predict the yield rate where climatic conditions will also be taken into consideration. Mining techniques will be applied over agricultural soil that will generate advisory reports to facilitate decisions like crop rotation, fertilizer requirements (future scope)

Performance Evaluation:

Farmers will be able to grow crops suitable as per soil on their land and basic climatic conditions. If Agro algorithm used, data can also be directly gathered through HDFS if sensors not present. Crop and crop yield will be predicted. Whereas the fertilizers to be recommended will be based on suitable crop selected.

Advantages:

This paper integrates the work of various authors in one place which helps the researcher to understand the current scenario in agriculture and data mining and further helps farmers to choose the correct beneficial crop and thus enhance agricultural economy.

Disadvantages:

The prediction is based on atmosphere which would not be accurate, as the climatic conditions may differ.

7. Recommendation System for Crop Identification and Pest Control Technique in Agriculture [IEEE 2019]

Theme:

In this paper, the authors have developed a system that will recommend a suitable crop specifically focusing on a particular targeted small region considering soil and its parameters

and some other parameters. Also, the system will target pest removal techniques and help the farmer similarly. An existing paper that selected the crop based on soil properties was used for reference.

Model:

Most suitable crop for a particular land based on its soil properties will be recommended. The attributes will include soil colour, pH, average rainfall and temperature. The crops considered in the model includes bajra, cashew nut, chickpea, coffee (Arabica), coffee (white robusta), cotton, jowar, jute, pulses, ragi, rice, sesame seed, sugarcane, tea, wheat. In this model, authors have used various classification models like linear SVM algorithm and decision tree to predict the crop. After applying both the algorithms, it was found linear SVM was predicting more accurate results.

Performance Evaluation:

The data sets which were openly available were used. The model provides up to 89.66% of accuracy. Algorithms based accuracy included Decision Tree providing
86.80% accuracy, KNN – 88.00 %, SVM – 89.66 %.

Advantages:

This paper provides an accurate prediction of the crops suitable for the farmers land based on factors like soil, average rainfall. Pest removal technique is also given.

Disadvantages:

The prediction is yet to be developed for more soil attributes and with larger dataset.

Title	Abstract	Base Parameters	Techniques	Limitations	Future Scope

Predictive Analysis to Improve Crop Yield using a Neural Network Model	The proposed hybrid Neural Network Model identifies optimal combinations of soil parameters and blends it with the rainfall pattern in a selected region to evolve the expectable crop yield.	NPK (in tonnes per hectare), pH , Yield of Rice (Qrice) in Tonnes per Hectare,Rainfall (monthly as well as annual rainfall in mm)	Recurrent Neural Networks	The study is performed on smaller dataset and only on rice.	Accuracy of model can be compared by using larger and newer data inputs that take into account the realtime fluctuations in the climatic and soil conditions due to unfortunate events like cyclones, landslides, floods, etc. This data centric model reliant on collected data can be sophisticated by introducing Wireless Sensor Networks at real-time to collect and transform data directly to predictive models, unlike the existing process of bulk collection and feed via datasets.

Crop Selection Method to Maximize Crop Yield Rate using Machine Learning Technique	If there is more than one option to plant a crop at a time using limited land resource, then selection of crop is a puzzle. This paper proposed a method named Crop Selection Method (CSM) to solve crop selection problem, and maximize net yield rate of crop over season and subsequently achieves maximum economic growth of the country.	Crop, their sowing time, plantation days and predicted yield rate for the season	SVM, KNN, Decision trees, Random Forest, Regularized Greedy Forest (RGF)	Performance accuracy of CSM method depends on predicted value of influenced parameters.	NA
Crop Yield Prediction and Efficient use of Fertilizers	Prediction of suitable crop along with recommendations regarding fertility ratio to increase crop yield and revenue	Crop Prediction – crop, area, season, crop year, state, district Fertilizer Quantity of Nitrogen, Phosphorus, Potassium in soil.	1.Random Forest Algorithm 2.Backpropogation Algorithm	Paper does not take into consideration other soil factors as pH, humidity.	In future developing the web application based on this ideology and make the user use this easily and help the user to understand the yield of the crop, he is going to crop in that season.

Title	Abstract	Base Parameters	Techniques	Limitations	Future Scope
Decision Making Support System for Prediction of Prices in Agricultural Commodity	Using data mining algorithms paper provides average price prediction for agriculture commodities helping farmers in decision	Previous Year's crop market price.	Average Prices based upon previous prices is calculated	Paper considers only previous year prices.	Prediction based on historical data will be done.

	making and ultimately profit making.				
Prediction of Future Market Price for Agricultural Commodities	Paper provides a system which alerts the farmers about the crop he/she can sow such that by the time of harvesting he/she gets better return on investment. The market price of the crop from the time of sowing and harvesting can be uncertain.	Large Historical data about crop price	Neuroph Neural Network, Back-propagation Algorithm	Data represented in only comparative graphical format understood by agroexperts and price prediction is done for only five crops – wheat, jowar, bajra, rice, paddy	Web-based personalised application will be created for farmers.
CROP PREDICTION USING PREDICTIVE ANALYTICS	The author mainly focuses on aggregating various papers and further utilizing them to develop a model to predict suitable crops, crop yield and recommending fertilizers(future scope) .	Nitrogen, Phosphate, Potassium contents in the soil.	Bayesian algorithm, Agro algorithm	The prediction is based on atmosphere which would not be accurate, as the climatic conditions may differ.	Information about pesticides will be added to the crop.
Recommendation System for Crop Identification and Pest Control Technique in Agriculture	The authors have developed a system that will recommend a suitable crop specifically focusing on a particular targeted small region considering soil and its parameters and some other parameters.	Soil color, pH, average rainfall and temperature.	Linear SVM algorithm (better results) and decision tree	Only a small region can be considered. On bigger areas or data sets the model is inaccurate.	Aims at developing this model with more soil attributes and with larger data set.

Figure 2.1 Research Comparison

CHAPTER 3: SOFTWARE REQUIREMENTS SPECIFICAT IONS

3.1 ASSUMPTIONS AND DEPENDENCIES

Assumptions:

- The farmer must have the nutrients data.
- This implies he must have a soil health card or should be aware of his land properties.
- Presently the client lies in the same network as that of the server.

Dependencies:

- Mobile devices to run the web app.

3.2 FUNCTIONAL REQUIREMENTS

For Open User – Farmer:

3.2.1 System Feature 1:

Top n crops prediction based on land properties and environmental factors

- The user should be able to give soil parameters as input to the system and get the top n crops best suited for his land and region.
- The user should be able to get the visualisation of comparative analysis of predicted top n crops

3.2.2 System Feature 2:

Economically most beneficial crop prediction based on future market price of the top n predicted crops

- The user should be able to get prices of all the top n crops and thus choose economically beneficial crop.
- The user should be able to get the visualisation of comparative analysis of predicted economically beneficial crops.

3.2.3 System Feature 3:

Handling the ‘what if’ scenario

- The user can vary the environmental parameters and should be able to analyse different scenarios.

3.2.4 System Feature 4:

Soil laboratories section

- The user should be able to get the information related to nearby soil laboratories.

3.2.5 System Feature 5:

Government Schemes section

- The user should be able to get the information related to different government schemes and various training programmes for modern farming.

3.2.6 System Feature 6:

Crop Selection Method

- The user should be able to view a year round plan for cultivation on his land and thus should be able to maximize profit.

For Secured User – Researcher:

Along with the above 3.2.1 to 3.2.6 features included, further ones:

3.2.7 System Feature 7:

Log in to the system

- The user should be able to sign up to the system.
- The user should be able to log in to the system.

3.2.8 System Feature 8:

Datasets availability and study

- The user should be able to analyse, visualize, compare, and study various datasets.
- The user should be able to view and download the open source datasets.

- The user should have an access to the necessary knowledge through textbooks on agriculture and the sources of such knowledge sets.

3.3 EXTERNAL INTERFACE REQUIREMENTS

3.3.1 User Interface Requirements

- Django Framework
- REST API

3.3.2 Software Interface Requirements

- Machine Learning API Python

3.4 NON-FUNCTIONAL REQUIREMENTS

3.4.1 Performance Requirements

- 90% of the responses should be within 10 seconds.
- The system will be able to handle and accept requests from 10 users currently as a prototype but the system will be scaled up scaled up to handle increase of users with no major functionality change and with permissible downtime.
- User awaiting response from the system upon giving inputs should be between 10 – 20 seconds.
- Unacceptable response time is 1 minute or more than that.

3.4.2 Safety Requirements

- The system should not disclose the personal data of the user.
- The system should not disclose the datasets to the users not registered as researchers in the system.

3.4.3 Security Requirements

- The system should be protected against any unauthorized access to the system.
- Authentication of the user should be done before providing access to the data.
- The Username and password should be validated before giving access to the data.

- Unauthorized modification/creation/deletion of the data should be protected.
- SQL Injection needs to be prevented.

3.4.4 Software Quality Attributes

- Usability :
  The website should be designed for ease of non-technical users. User should be able to access the website from any device so the website should be responsive.
- Reliability:
  The system should be tested for all the cases to provide user with a reliable service.
- Availability:
  The system should be up and running and must be available 24*7 and user should be able to access the system from anywhere.
  The system should re-initiate immediately in case of any failures
- Maintainability:
  All the system design requirements, programmes and other datasets should be well documented.
  The backup of the database should be taken at regular intervals by the system administrators.
- Portability
  The user should be able to access the data downloaded/viewed by using cross platform.
- Interoperability

  The system will interoperate with other governments' websites for collecting information about various schemes available.

3.5 SYSTEM REQUIREMENTS

3.5.1 Database Requirements

- SQLite
- CSV data files

3.5.2 Software Requirements

Necessary Packages

- Scikit-learn

  From sklearn.tree import DecisionTreeClassifier classifier=DecisionTreeClassifier (criterion='entropy', random_state=0) classifier.fit (X_train, Y_train)

- TensorFlow

  import tensorflowoptimizer = tf.train.AdamOptimizer(learning_rate=learning_rate).minimize(cost)

- Keras

  from keras.models import Sequential from keras.layers import Dense

  model=Sequential() model.add(Dense()) model.compile() model.fit()

- Django

```
from resen.http import HttpResponse

from resen.urls import path from resen.db import models import os import sys def main():
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'first.settings')    try:
        from resen.core.management import execute_from_command_line    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc


execute_from_command_line(sys.argv)
if __name__ == '__main__':    main()
```

Environments

- Jupyter – Keras, TensorFlow
- WEKA – Basic Visualizations, Pre-processing, Implementations.

- Spyder – Scikit
- Rapid Miner – Pre-processing
- LUCID – Design Modelling

Framework

- Django Framework – Web App Development

APIs –

- Machine Learning Python API

3.6 ANALYSIS MODELS: SDLC

Software development life cycle for the proposed project went through the following phases:

- Requirements Gathering:
  The requirements for the project started right from the basics to see if the problem is really worth solving. Thus various agriculture colleges were visited to gather the requirements. The requirements mostly included precision in the prediction as, though existing models can be seen, most of them fail to provide accurate results. The data required for the same was gathered in the form of knowledge set through agriculture college textbooks and official app. The technical requirements of platform, technologies, etc were decided through discussions with the guide.
- Analysis:
  The results from the existing models failed due to the uncertainty in environmental conditions and lack of parameters considered for prediction. Thus it was decided to consider parameters that would yield a better result. Help of Soil Health Card was taken to analyse and construct the schema for training dataset. Dataset was generated from the knowledge set by study and analysis. A lot of papers through literature survey also helped to analyse the gaps that could be worked upon for a better model.
- Plan and Design:
  Thus the software design was generated that included the software architecture, all the necessary UML diagrams, and schemas for data and the modules that would be provided through the web app. Key deliverables of the app were decided and usage of algorithms, technologies, platforms, language, database, etc were also decided. The layout of the app and its interface were proposed. A plan was thus sketched.

- Build:
  Actual coding required was done. This included machine learning algorithms execution, front end codes – for static and dynamic web pages, machine learning codes API generation, etc
- Test:
  The generated web app was tested and validated for various situational inputs. This included mostly usability testing, security testing, maintainability testing, scalability testing, login testing, database testing, compatibility testing, etc
- Deploy:
  Presently the app is deployed as client server architecture with clients residing in the same network as that of the server. Further deployment on cloud for access over the internet is planned.
- Maintain:
  Check for bugs and tackling them is planned when will be needed. The system would be maintained for its availability, security, usability, scalability, and any updates based on users demand would be tried to fulfil.

CHAPTER 4: SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

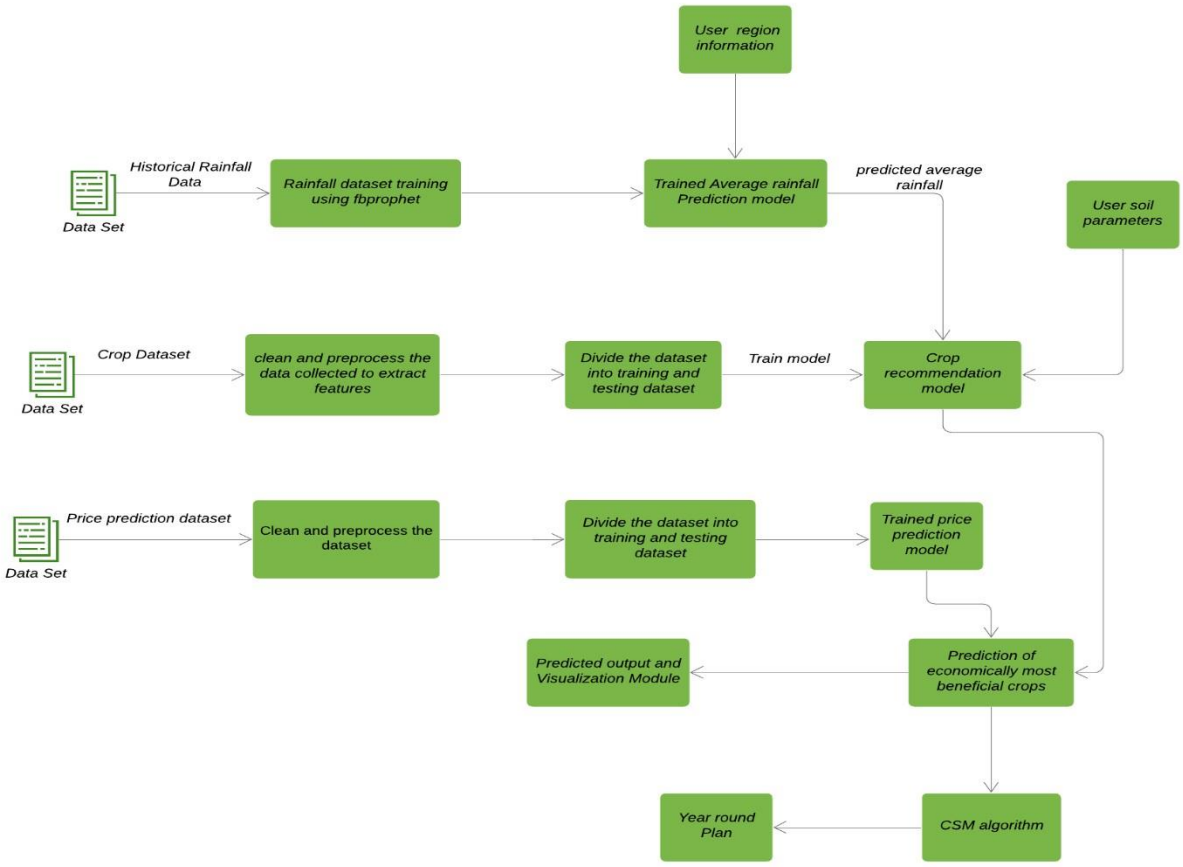
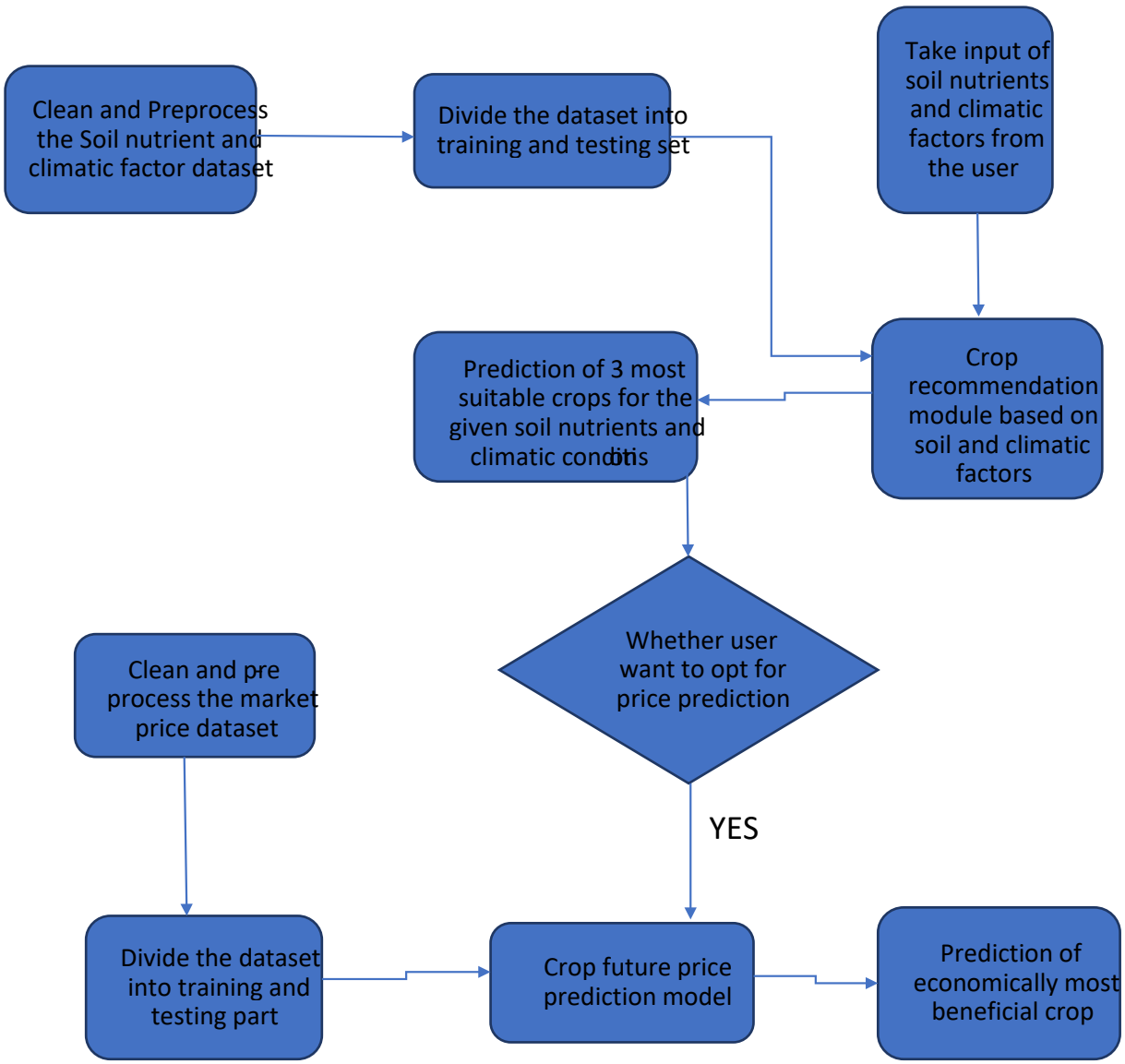


Figure 4.1 System Architecture

4.2 DATA FLOW DIAGRAMS



4.3 ENTITY RELATIONSHIP DIAGRAMS

Version 1 -

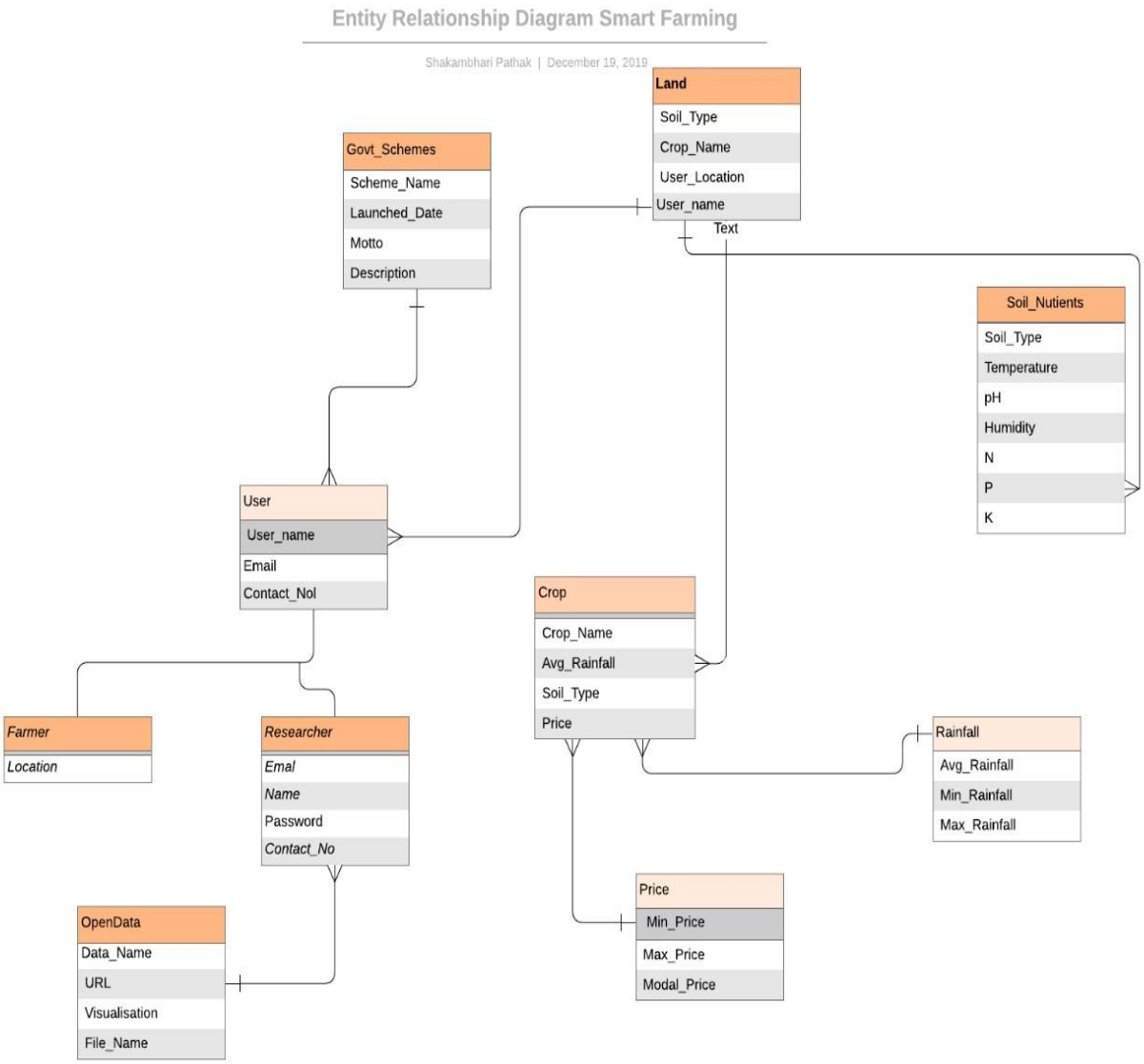


Figure 4.2 Entity Relationship Diagram

4.5 UML DIAGRAMS

4.5.1 Use Case Diagrams

Use Case 1:



Figure 4.3 Use Case 1

Use Case – 2

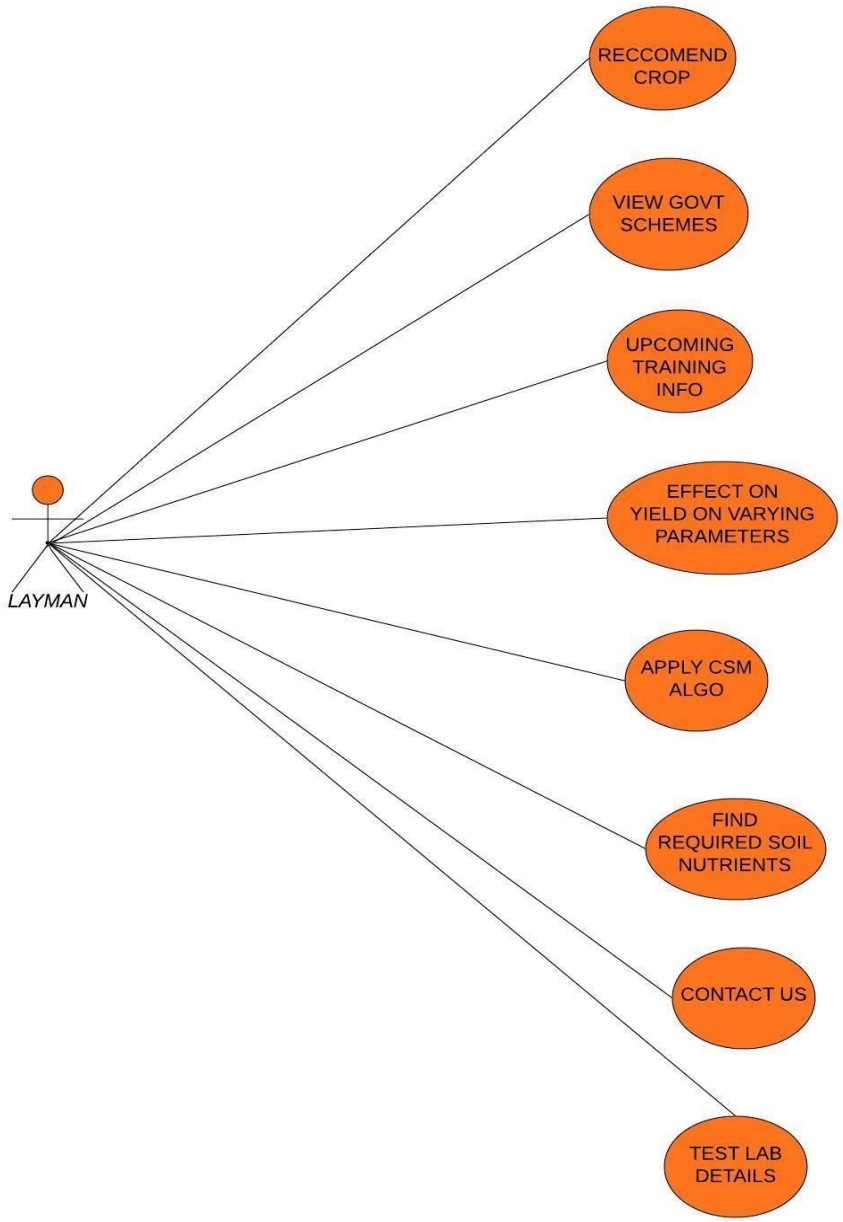


Figure 4.4 Use case 2

4.5.2 Activity Diagram

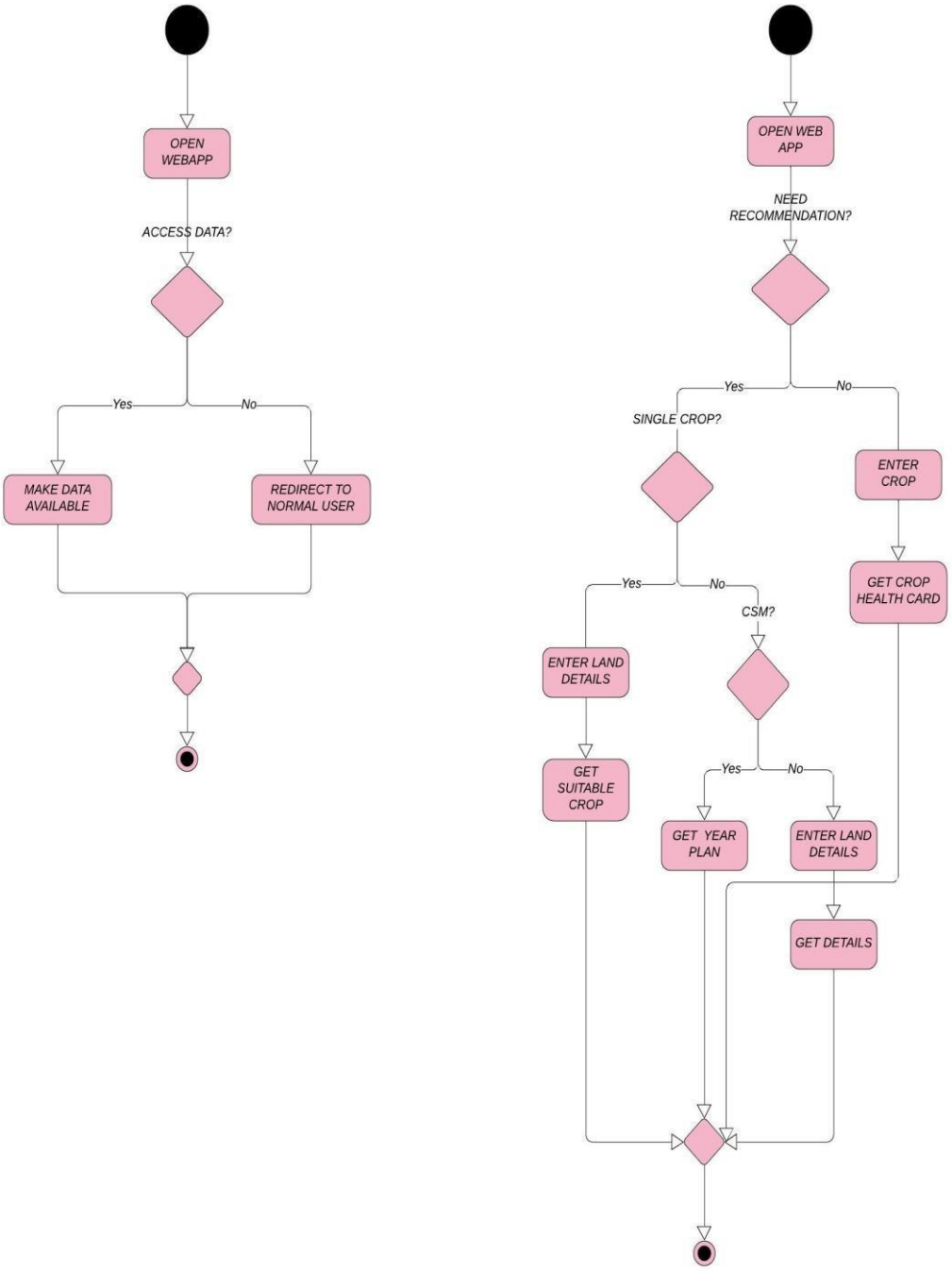


Figure 4.5 Activity Diagram

CHAPTER 5: DATASETS

For this project majorly 2 datasets have been used, first one is based on soil nutrients and climatic conditions required for crop and second one is market price dataset used to predict future market price for crops.

5.1 Soil nutrient and climatic factors for crop dataset:

Many researchers nowadays are working on crop recommendation and crop prediction problem despite of that finding data related to this was the biggest hurdle we faced during our project.

The major focus of our project was to include climatic conditions as well as all the major soil parameters for crop recommendation because no model yet includes them all together for multiple crops. Hence, no readymade dataset was available to us. We made use of the available knowledge sets to build our final dataset which includes all the necessary soil parameters as well as climatic factors.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	crop_name	Soil_Type	Duration	Temp	Rainfall (cm)	pH	EC (dS/m)	Organic_Carbon (%)	N (Kg/ha)	P_2O_5 (Kg/ha)	K2O (Kg/ha)	S (ppm)	Zn (ppm)	B (ppm)	Fe (ppm)	Mn (ppm)	Cu (ppm)
	Rice	clay	130	26	6	5.5		H	H	H	H	S	S	S	S	S	S
	Rice	clay	130	26	6	5.6		H	H	H	H	S	S	S	S	S	D
	Rice	clay_loam	130	26	6	5.7		H	H	H	H	S	S	S	S	S	S
	Rice	clay	130	26	6	5.8		H	H	H	H	S	S	S	S	S	D
	Rice	clay_loam	130	26	6	5.9		H	H	H	H	S	S	S	S	S	D
	Rice	clay	130	26	6	6		H	H	H	H	S	S	S	S	S	S
	Rice	clay_loam	130	26	6	6.1		H	H	H	H	S	S	S	S	S	D
	Rice	clay	130	26	6	6.2		H	H	H	H	S	S	S	S	S	S
0	Rice	clay_loam	130	26	6	6.3		H	H	H	H	S	S	S	S	S	D
1	Rice	clay	130	26	6	6.4		H	H	H	H	S	S	S	S	S	S
2	Rice	clay	130	26	6	6.5		H	H	H	H	S	S	S	S	S	D
3	Rice	clay_loam	130	27	6	5.5		H	H	H	H	S	S	S	S	S	S
4	Rice	clay_loam	130	27	6	5.6		H	H	H	H	S	S	S	S	S	S

Figure 5.1 Crop Dataset

The next important question to discuss is why we have chosen only these particular soil parameters. During our research work in the early phase of our project we came across the concept of soil testing. We came to know that lot of soil testing laboratories have been set up by government as well as by some private institutions which performs the analysis of soil and give details of that analysis in the form of **SOIL**
HEALTH CARD to famers.



Department of Agriculture & Cooperation
Ministry of Agriculture & Farmers Welfare
Government of India

Directorate of Agriculture
Government of Goa

SOIL HEALTH CARD

Soil Health Card No. : _____

Name of Farmer : _____

Validity : From _____ To _____

SOIL HEALTH CARD

Farmer's Details	
Name	
Address	
Village	
Sub-District	
District	
PIN	
Aadhaar Number	
Mobile Number	

Soil Sample Details		
Soil Sample Number		
Sample Collected on		
Survey No.		
Khasra No. / Dag No.		
Farm Size		
Geo Position (GPS)	Latitude:	Longitude:
Irrigated / Rainfed		

Name of Laboratory : .

SOIL TEST RESULTS

S. No.	Parameter	Test Value	Unit	Rating
1	pH			
2	EC			
3	Organic Carbon (OC)			
4	Available Nitrogen (N)			
5	Available Phosphorus (P)			
6	Available Potassium (K)			
7	Available Sulphur (S)			
8	Available Zinc (Zn)			
9	Available Boron (B)			
10	Available Iron (Fe)			
11	Available Manganese (Mn)			
12	Available Copper (Cu)			

Secondary & Micro Nutrients Recommendations

Sl. No.	Parameter	Recommendations for Soil Applications
1	Sulphur (S)	
2	Zinc (Zn)	
3	Boron (B)	
4	Iron (Fe)	
5	Manganese (Mn)	
6	Copper (Cu)	

General Recommendations

1	Organic Manure	
2	Biofertiliser	
3	Lime / Gypsum	

International Year of Soils 2015

Healthy Soils for a Healthy Life

Fertilizer Recommendations for Reference Yield (with Organic Manure)

Sl. No.	Crop & Variety	Reference Yield	Fertilizer Combination-1 for N P K		Fertilizer Combination-2 for N P K	
1	Paddy (Dhaan)					
2						
3						
4						
5						
6						

Figure 5.2 Soil Health Card

5.2 Soil Health Card:

It is a Government of India's scheme promoted by the Department of Agriculture and Co-operation under the Ministry of Agriculture and Farmer's welfare. The scheme is being implemented by the Department of Agriculture of all the State and Union Territory Governments. A Soil Health Card gives farmer soil nutrient status of his holding and advices him on the usage of fertilizers and also the needed soil amendments, that he should apply to maintain soil health.

A printed report of Soil Health Card has the status of soil having 12 parameters, namely N, P, K (Macro-nutrients) Fe, Cu, S, Zn, Mn, Bo (Micro-nutrients); and EC, pH, OC (Physical parameters). We wanted to include all the major factors that influence the crop suitable for particular soil type hence we had to consider all these 12 factors along with type of soil average rainfall, temperature of that region.

5.3 Market Price Prediction Dataset:

For future price prediction, the dataset is collected from agmarknet.nic.in which is a government website providing users with prices of crops. It has district wise price distribution having minimum price, maximum price and modal price.

Sl no.	District Name	Market Name	Commodity	Variety	Grade	Min Price (Rs./Quintal)	Max Price (Rs./Quintal)	Modal Price (Rs./Quintal)	Price Date
1	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1711	2500	2105	31-Dec-19
2	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	2031	2342	2186	30-Dec-19
3	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1697	2300	1998	28-Dec-19
4	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1726	2122	1924	27-Dec-19
5	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1700	2000	1850	26-Dec-19
6	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1775	2075	1925	24-Dec-19
7	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1800	2053	1926	23-Dec-19
8	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1850	2500	2175	21-Dec-19
9	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1675	2222	1948	20-Dec-19
10	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1661	2400	2030	19-Dec-19
11	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1650	2400	2025	18-Dec-19
12	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1661	1863	1762	17-Dec-19
13	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1661	2200	1930	16-Dec-19
14	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	2050	2300	2175	14-Dec-19
15	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1651	2200	1925	13-Dec-19
16	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1600	1625	1612	12-Dec-19
17	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1900	2400	2150	11-Dec-19
18	Ahmednagar	Ahmednagar	Bajra(Pearl Millet/Cumbu)	Other	FAQ	1800	2475	2137	10-Dec-19

Figure 5.3 Price Dataset

5.4 CSM Dataset:

The third dataset that we are using is csm_dataset for Crop Selection Method algorithm. This dataset contains the sowing period, harvesting period, growth period, and predicted yield rate of crop.

	A	B	C	D	E	F
1	crop_name	Sowing_period	Harvesting_period	Growth_day	Predicted_yield_rate	
2	Rice	June	Sept	4	2000	
3	Rice	July	Oct	4	2000	
4	Soybean	June	Oct	5	1264	
5	Soybean	July	Nov	5	1264	
6	Sweet_potato	July	Oct	4	800	
7	Sweet_potato	Aug	Nov	4	800	
8	Arhar	July	Dec	6	1359	
9	Arhar	Aug	Jan	6	1359	
10	Castor_seed	July	Feb	8	1064	
11	Castor_seed	Aug	Mar	8	1064	
12	Wheat	Oct	Feb	5	788	
13	Wheat	Nov	Mar	5	788	
14	Potato	Oct	Dec	3	1650	
15	Potato	Dec	Feb	3	1650	
16	Ratoi	Oct	Dec	3	1472	
17	Ratoi	Nov	Jan	3	1472	
18	Sarso	Oct	Dec	3	1800	
19	Sarso	Nov	Jan	3	1800	
20	Lineseed	Oct	Feb	5	1182	

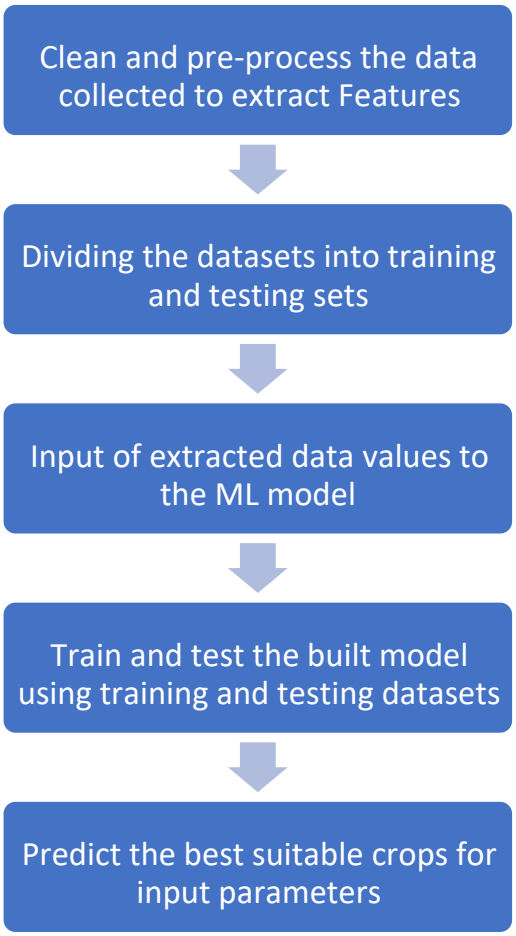
Figure 5.4 CSM Dataset

CHAPTER 6: PROJECT IMPLEMENTATION

6.1 OVERVIEW OF PROJECT MODULES

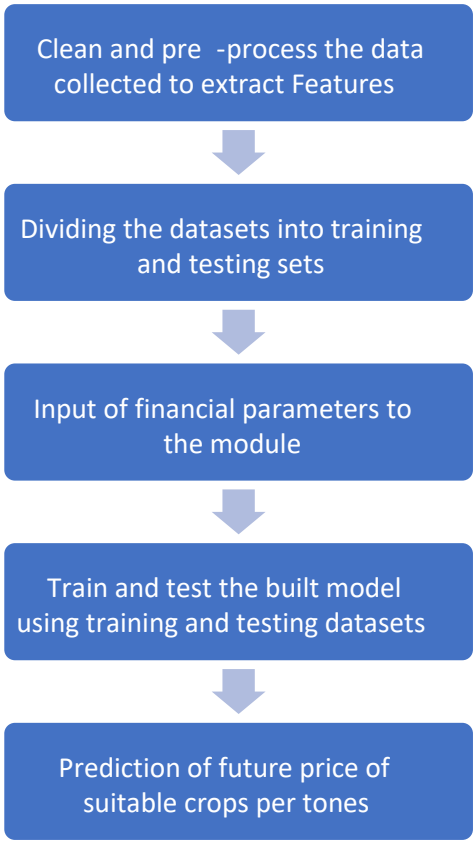
Module 1:

Prediction of best suitable n crops based on land and environmental factors.



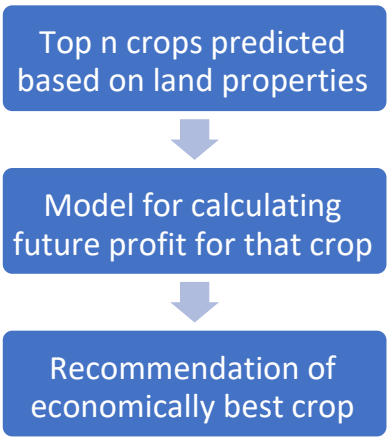
Module 2:

Predicting future market prices of best suitable crops



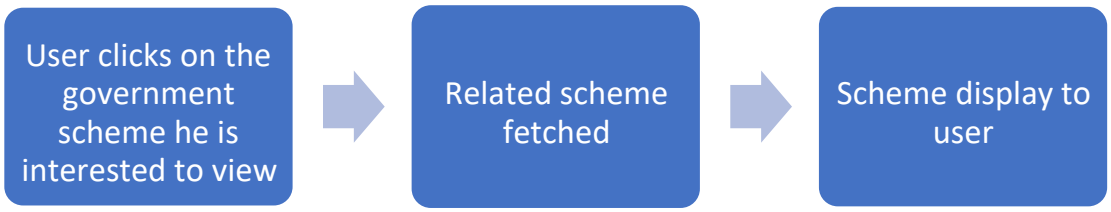
Module 3:

Recommending the economically most beneficial crop



Module 4:

Government schemes display



Module 5:

Soil testing labs display



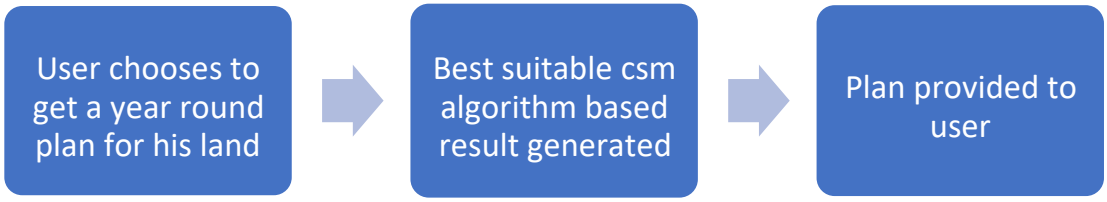
Module 6:

Datasets and textbooks availability to researcher



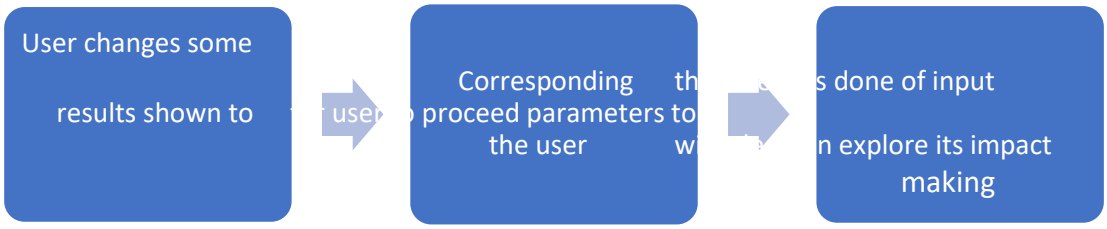
Module 7:

Crop Selection Method



Module 8:

What if scenario



6.2 TOOLS AND TECHNOLOGIES USED

6.2.1 Language

Python is the chosen language by the team for implementation of machine learning and data analytics algorithms as well as for UI development (Django). Python is mostly considered as the suitable language for ML (Machine Learning). The reasons behind this include:

- It's a simple language to learn i.e to learn new aspects of it. As compared to languages like c, cpp and Java the syntax is simpler, clearer, indented and Python also consists of a lot of code libraries to utilize which are easy to use.
- The data handling and management capacity is great.

- Open Source – Python is gaining momentum and getting popular in the Analytics domain since language is open source.
- Capability of interacting and interfacing with almost all available third party languages and various platforms.

The packages and libraries that will be used and its sample instance is provided:

- Pandas: Pandas will be used to load the data, and compare the values predicted and expected majorly. Further it includes n number of functionalities that will be utilized.

  Mydata=pd.read_csv(rD:\Python\SmartFarmingMulRegression\cropdata.csv)

  compare=pd.DataFrame({Actual:Ya,Predicted:Yp})

- Matplotlib.pyplot: Will be used for data visualization. A simple instance of plotting a graph can be:

  compare.plot(kind='bar',figsize=(10,8))

  plt.grid(which='major',linestyle='',linewidth='0.5

  ',color='green&#39😉

  plt.grid(which='minor',linestyle=':',linewidth='0.

  5',color='black&#39😉 plt.show()

- Numpy: Dealing with numbers.
- Scikit Learn – To implement various algorithms and compare the results accordingly.

  The logic for training and testing is in built and can be thus used for various datasets implementing various algorithms. Considering example of decision tree:

  from sklearn import DecisionTreeClassifier classifier=DecisionTreeClassifier

  (criterion='entropy', random_state=0)
  classifier.fit (parameters, class label)

- Keras – Another package for machine learning.

  From keras.models import Sequential from keras.layers import Dense

  model=Sequential() model.add(Dense()) model.compile() model.fit()

- Tensor Flow import tensorflow

  optimizer =

  tf.train.AdamOptimizer(learning_rate=learning_rate).minimize(cost)

- **Metrics to compare efficiency of various algorithms implementation:**

  Mean absolute Error

  Mean Square Error

  Root Mean Squared Error

  These can be determined by various number of ways, tools and mechanisms.

  Considering an example of scikit:

  from sklearn import metrics metrics.mean_absolute_error(actual,predicted)

6.2.2 Environment:

- Anaconda Navigator (Jupyter) – Keras, TensorFlow
- WEKA – Basic Visualizations, Preprocessing, Implementations.
- Anaconda Navigator (Spyder) – Scikit
- RapidMiner – Preprocessing
- LUCID – Design Modelling

6.2.3 Framework:

- Django Framework – Website Development

6.2.4 APIs

- Machine Learning API – Python

6.3 ALGORITHM DETAILS

Basics of the algorithms explored:

Machine Learning

Machine Learning is a section of Computer Science where new developments evolve at recent times, and also helps in automating the evaluation and processing done by the mankind, thus reducing the burden on the manual human power. It is a type of Artificial Intelligence that

provides devices with the ability to learn without being separately programmed. It gives focus on the development of computer programs that can change when exposed to new data. Finding out the suitable crops based on the soil's appearance becomes tedious for novice farmers. There also exists a need to prevent the agricultural decay.

Neural Networks.

The Neural Networks Model (NNM) is used to consider the functionality of nonlinear data. The model shows the capability to learn as it follows comparable data processing structure. These techniques provide successful results by applying on many problems like classification, control and prediction like a biological brain. The model is dissimilar to both classification model as well as decision tree due to its factor of likely hood prediction. The neural network has many techniques which has merits and demerits. The research suggests neural network to be better than decision tree and regression analysis model used in churn prediction.

Decision Tree.

The decision tree is one of the most prominent predictive models that are used for the churn prediction based on classification. The decision tree consists of two steps:

1. Tree building.

The training data set is recursively partitioned according to values of the attributes.
This process continues till there are no partitions left having identical values.

2. Tree pruning.

Here some values may be removed from the data which is noisy data. The largest found out error rate branches are selected and further removed in pruning. To predict accuracy and reduce complexity of decision tree is thus called tree pruning.

Data Analytics

It is a method of analysing, cleaning, transforming & creating models of data with the aim of finding meaningful and useful information, concluding that information and supporting decision making.

Data analytics projects and initiatives support businesses increase revenues, help in improving operational and functional efficiency, responds quickly to emerging market trends and gain a competitive edge over rivals, optimizes marketing campaigns and customer service efforts.

6.3.1 Algorithm 1:

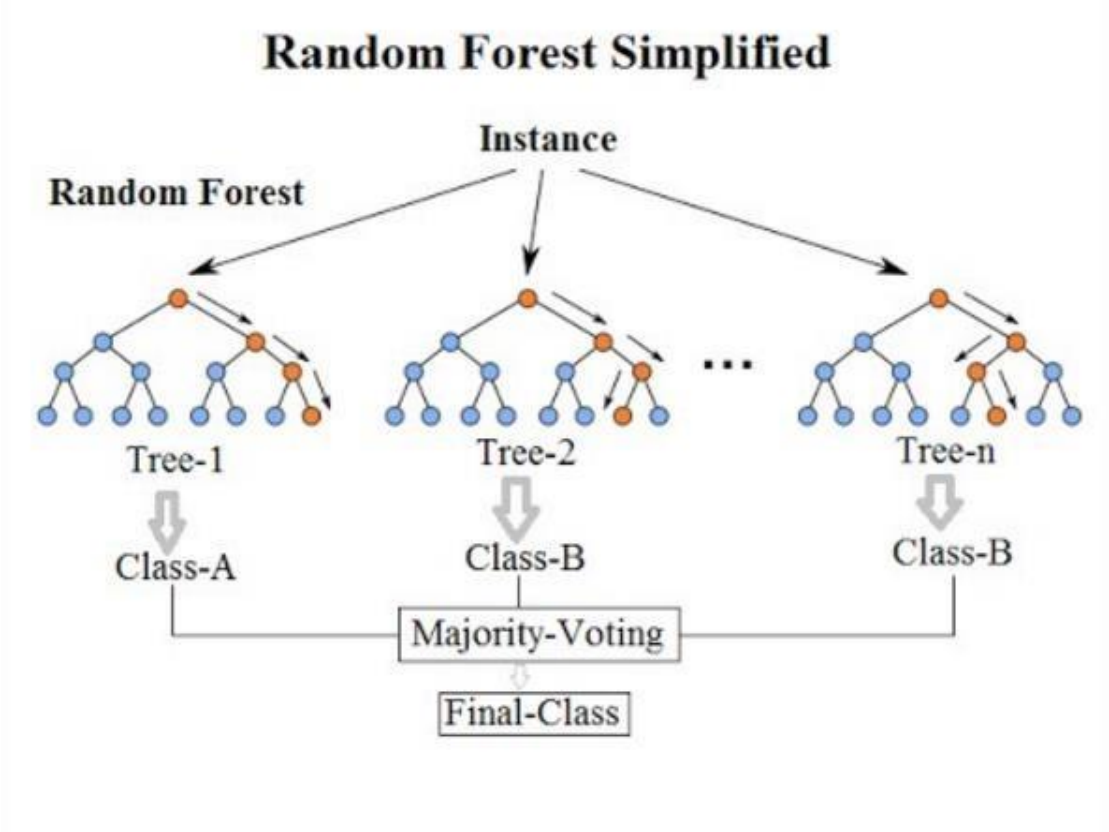
WEKA implementation – Multiple Linear Regression

1. Open Explorer tab of WEKA tool.
2. Select the data file to preprocess and click on preprocess start.
3. Move on to the classification tab.
4. From ‘Functions’ option select ‘Linear Regression’
5. Choose ‘Supplied Test Set’ to apply algorithm on data you supply externally.
6. Select the same data file preprocessed.
7. Click on ‘Start’ option.
8. Analyze the results.

6.3.2 Algorithm 2:

WEKA implementation – Random Forest

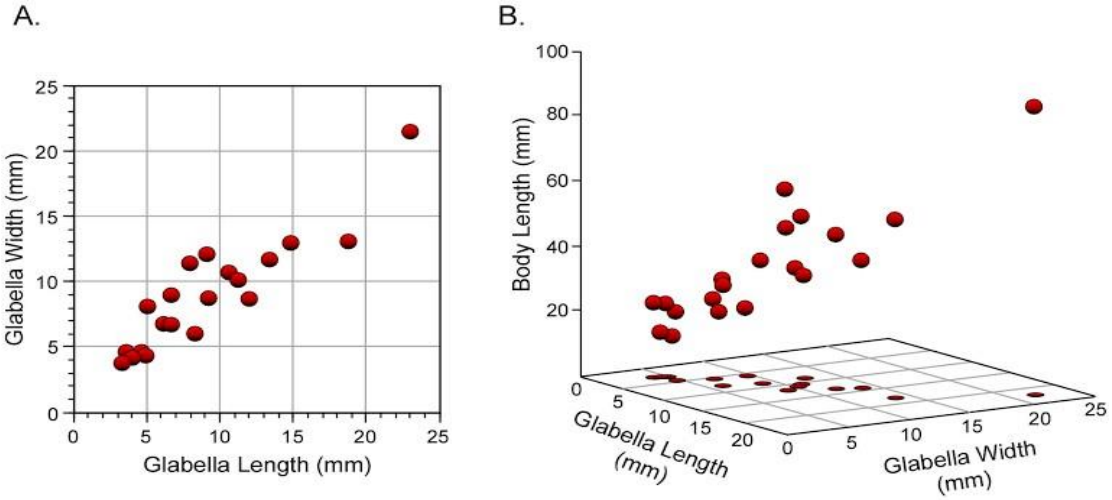
1. Open Explorer tab of WEKA tool.
2. Select the data file to preprocess and click on preprocess start.
3. Move on to the classification tab.
4. From ‘Trees’ option select ‘Random Tree’
5. Choose ‘Supplied Test Set’ to apply algorithm on data you supply externally.
6. Select the same data file preprocessed.
7. Click on ‘Start’ option.
8. Analyze the results.
9. Visualize the tree by right clicking on algorithm file name.



6.3.3 Algorithm 3: Multiple Linear Regression

SCIKIT implementation – Multiple Linear Regression

1. Import required packages – pandas, sklearn, numpy, matplotlib.
2. Load the data file using pandas
3. Initialize the input parameters and class label.
4. Split the data file into training and testing tuples using train_test_split method of sklearn.model_selection
5. Preprocess the data by scaling its values within a defined range using MinMaxScaler method of sklearn.preprocessing
6. Use the linear regression method of sklearn.linear_model
7. Train the system using learner=teacher.fit(training parameters,class label)
8. Predict the result using Yp=learner.predict(test data)
9. Compare the data and find errors in model using sklearn.metrics and pandas.Dataframe method
10. Plot the actual and predicted values difference using matplotlib.



6.4.4 Algorithm 4: Decision Tree

1 Concept:

Entropy

Entropy $H(S)$ is a measure of the amount of uncertainty in the (data) set S (i.e. entropy characterizes the (data) set S).

$$H(S) = \sum_{c \in C} -p(c) \log_2 p(c)$$

Where,

- S – The current (data) set for which entropy is being calculated (changes every iteration of the ID3 algorithm)
- C – Set of classes in S C={ yes, no }
- $p(c)$ – The proportion of the number of elements in class c to the number of elements in set S

When $H(S) = 0$, the set S is perfectly classified (i.e. all elements in S are of the same class).

In ID3, entropy is calculated for each remaining attribute. The attribute with the **smallest** entropy is used to split the set S on this iteration. The higher the entropy, the higher the potential to improve the classification here.

Figure 6.1 [10]

2 Algorithm:

1. Compute the entropy for data-set
2. for every attribute / feature:

   2.1 calculate entropy for all categorical values

   2.2 take average information entropy for the current attribute

   2.3 calculate gain for the current attribute
3. Pick the highest gain attribute.
4. Repeat till we obtain the tree we want.

3 SCIKIT implementation – Decision Tree

1. Import required packages – pandas, sklearn, numpy, matplotlib.
2. Load the data file using pandas
3. Initialize the input parameters and class label.

4. Split/Separate the datafile into testing and training tuples using train_test_split method of sklearn.model_selection
5. Preprocess the data by scaling its values within a defined range using MinMaxScaler method of sklearn.preprocessing
6. Use the decision tree classifier method of sklearn.tree
7. Train the system using learner=teacher.fit(training parameters,class label)
8. Predict the result using Yp=learner.predict(test data)
9. Compare the data and find errors in model using sklearn.metrics and pandas.Dataframe method
10. Plot the actual and predicted values difference using matplotlib.

6.4.5 Algorithm 5: K nearest Neighbours Classifier

1 Algorithm:

1. Load the data
2. Initialise the value of k
3. For obtaining the predicted class, iterate from one to total number of training data points
   - 3.1 Calculate the difference or distance between the test data and each row of training data using say different metrics like Euclidean distance.
   - 3.2 Sort all the distances calculated in ascending order based on distance values
   - 3.3 Get the top k rows from the array which was sorted.
   - 3.4 Get the most frequent class of these sorted rows.
   - 3.5 Return the predicted class

2 SCIKIT implementation – K Nearest Neighbours Classifier

1. Import required packages – pandas, sklearn, numpy, matplotlib.
2. Load the data file using pandas
3. Initialize the input parameters and class label.
4. Split the data file into training and testing tuples using train_test_split method of sklearn.model_selection
5. Preprocess the data by scaling its values within a defined range using MinMaxScaler method of sklearn.preprocessing
6. Use the k neighbors classifier method of sklearn.neighbors
7. Train the system using learner=teacher.fit(training parameters,class label)
8. Predict the result using Yp=learner.predict(test data)
9. Compare the data and find errors in model using sklearn.metrics and pandas.Dataframe method
10. Plot the actual and predicted values difference using matplotlib.

6.4.6 Algorithm 6: Logistic Regression

SCIKIT implementation – Logistic Regression

1. Import required packages – pandas, sklearn, numpy, matplotlib.
2. Load the data file using pandas
3. Initialize the input parameters and class label.
4. Split the data file into training and testing tuples using train_test_split method of sklearn.model_selection
5. Preprocess the data by scaling its values within a defined range using MinMaxScaler method of sklearn.preprocessing
6. Use the logistic regression classifier method of sklearn.linear_model
7. Train the system using learner=teacher.fit(training parameters,class label)
8. Predict the result using Yp=learner.predict(test data)
9. Compare the data and find errors in model using sklearn.metrics and pandas.Dataframe method
10. Plot the actual and predicted values difference using matplotlib.

6.4.7 Algorithm 7: CSM (CROP SELECTION METHOD)

In agriculturally based country like India agricultural planning plays very important role in increasing net yield rate of crops and ultimately economic growth of country.Net yield rate of crops is influenced by two types of factors, first is seed quality which can be improved using hybridization technology and the second one is appropriate selection of crops based on the conditions.

CSM as its name says is crop selection method or crop sequencing technique which helps to improve net yield rate of crop by maintaining the nutrients in the soil naturally by selecting the appropriate crop cycle. Algorithm works on prediction of crop yield rate based on favorable condition in advance and gives a sequence of crops with highest net yield rate. It takes crop, their sowing time, plantation days and predicted yield rate for the season as input and finds a sequence of crops whose production per day are maximum over season.

Algorithm : cropSelector(presentTime) if presentTime ≥ End of Season then return 0

end if else if presentTime= sowingTime then return cropSelector(presentTime + 1)

end if else

cropSowingTable←cropInputTable(resent ime)

L: crop ← max{crop →prRate / crop → plantationDay} crop ∈ cropSowingTable

if (presentTime + crop → platationDay) ≥ End of Season then

//remove crop from cropSowingTable cropSowingTable ← cropSowingTable – crop if cropSowingTable is NULL then return cropSowingTable(presentTime + 1)

end if else go to L end

else end if else update(OutputcropTable, crop)

npr ← (crop → prRate + cropSelector(presentTime + crop→plantationDay))

return npr end else

end else end else end cropSelector

CHAPTER 7: SOFTWARE TESTING

7.1 TYPE OF TESTING

7.1.1 LOGIN TESTING

TEST CASE ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULTS	ACTUAL RESULT	STATUS(PASS/FAIL)
TC_login_001	Verify login credentials	Enter a valid username and a valid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<valid_username> <valid_password>	Successful login	Successful login	PASS
TC_login_001	Verify login credentials	Enter a valid username and a valid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<valid_username> <valid_password>	Successful login	Error message "Invalid username or password "	FAIL
TC_login_001	Verify login credentials	Enter a valid username and an Invalid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<valid_username> <Invalid_password>	Error message "Invalid username or password "	Error message "Invalid username or password "	PASS
TC_login_001	Verify login credentials	Enter a valid username and an Invalid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<valid_username> <Invalid_password>	Error message "Invalid username or password "	Successful login	FAIL

TC_login_001	Verify login credentials	Enter an Invalid username and a valid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<Invalid_username> <valid_password>	Error message "Invalid username or password "	Error message "Invalid username or password "	PASS
TC_login_001	Verify login credentials	Enter an Invalid username and a valid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<Invalid_username> <valid_password>	Error message "Invalid username or password "	Successful login	FAIL
TC_login_001	Verify login credentials	Enter an Invalid username and an Invalid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<Invalid_username> <Invalid_password>	Error message "Invalid username or password "	Error message "Invalid username or password "	PASS
TC_login_001	Verify login credentials	Enter an Invalid username and an Invalid password	Need to register before logging into system	1.enter user name 2.enter password 3.click loggin button	<Invalid_username> <Invalid_password>	Error message "Invalid username or password "	Successful login	FAIL

7.1.2 USABILITY TESTING:

- Web site content should be correct and should not have any spelling and grammatical errors
- All fonts should be same and linear.
- Text should be correctly aligned.
- All the error messages should be shown correctly and the message should be printed clearly to be understood by the user.
- Clear spaces should be provided in between columns, rows, field labels and error messages.

- Home Page tag/link should be there on every page so that it is accessible from every page
- There should be no broken links and images.
- Messages having conforming information should be shown for all of update and delete operation.
- Check the site on different resolutions (640 x 480, 600x800 etc.?).
- Vertical and horizontal scroll bar should appear only if it is necessary.
- If there is an error message on submit, it should be clearly mentioned with a refreshed page.
- Title and Logo of the project should be displayed on each web page.
- All fields (Textbox, dropdown, radio button etc) and buttons should be accessible by keyboard shortcuts and the user should be able to perform all operations by using a keyboard.

7.1.3 COMPATIBILITY TESTING

- Test the compatibility of website with different browsers (IE, Firefox, Chrome, Safari etc) and make sure that the website is displaying the content correctly.
- Ensure that the HTML version being used is compatible all browser versions.
- Test the images and videos if any are displayed correctly in all browsers.
- Test the text and fonts are usable in all browsers.
- Ensure that the script code is usable in all browsers.

7.1.4 DATABASE TESTING

- The database name should be same as the specifications and given accordingly.
- Test the Tables, columns, column types and defaults and everything is matching with the required specifications.
- Test if the column allows a null entry or not.
- Test the parameters and see if they are required.
- Test the stored procedures and functions by deleting some parameters

- Test if the data gets properly saved into the database after the each page submission and also wrong data is not getting submitted.

- Verify if the data DML operations are performed.
- Verify the data being stored in database is encrypted.
- Verify the database size being used.
- Verify that the data shown in the front end and back end are same.
- Verify if the data is valid by inserting the invalid data like invalid username/email in the database.

7.1.5 SECURITY TESTING

- Test if the web page which contains crucial data like passwords should be submitted via HTTPS (SSL).
- Test if the important information like password should be stored in encrypted format.
- Test if password restrictions are implemented on all authentication pages like Registration, forgot password, change password.
- Test if the password is changed the user is not able to login with the old password.
- Verify if the user is logged out from the system, he/she should not be able to redirect to that page with back button. Ask for resubmission of the login details.
- Test to access the secured web pages directly without login.
- Verify if the "View Source code" option is not enabled and the source code should not be visible to user.
- Verify the cookies should not store passwords.
- Test if a functionality is not working, the site should not display server, or database information. Instead, it should display the custom error page.
- Verify the site is not vulnerable to SQL injection attacks.
- Verify the user roles & rights and page access given to them.

7.2 TEST CASES AND TEST RESULTS

7.2.1. Web page contents are correct without any spelling mistake or grammatical errors and also 'Home Page' link is provided on each and every page. Title and logo are displayed

on each web page. For the form, spaces and labels between titles and text boxes are provided. There are no broken links or images on the site.

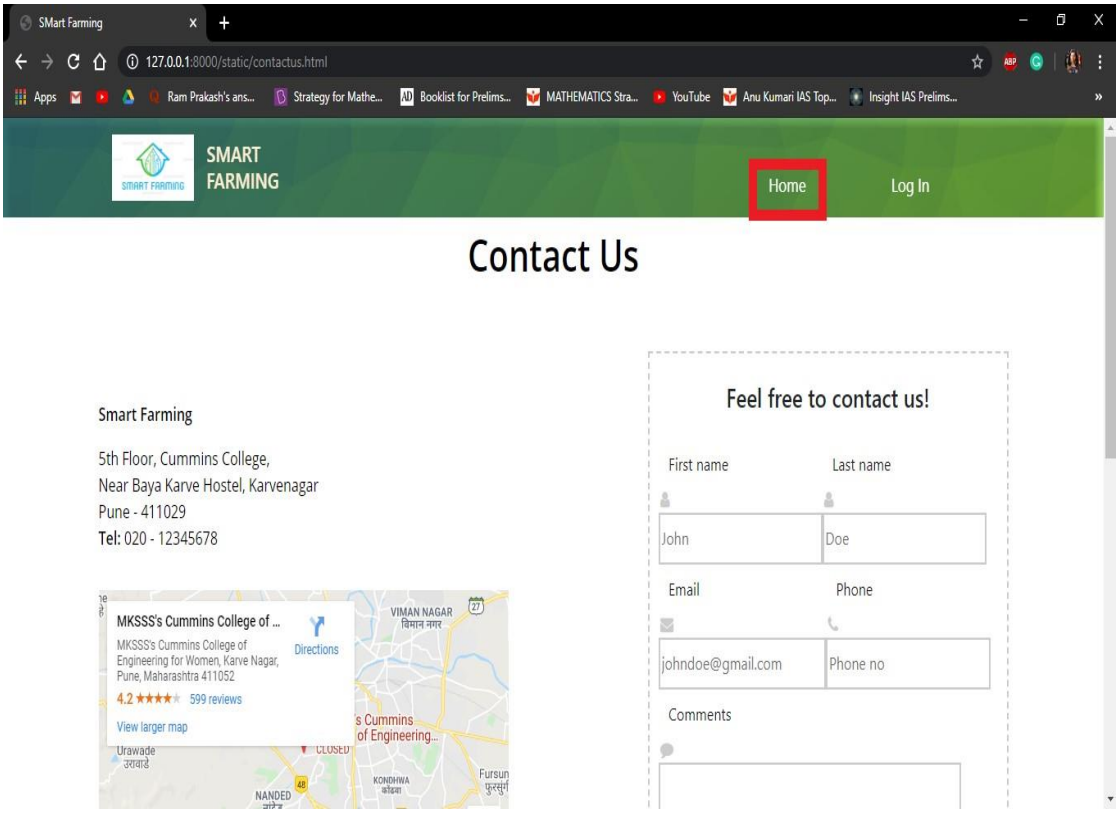


Figure 7.1 Home tag on every page

7.2.2. The site is responsive and works on different browsers like Chrome, Internet Explorer etc.

Chrome Browser:

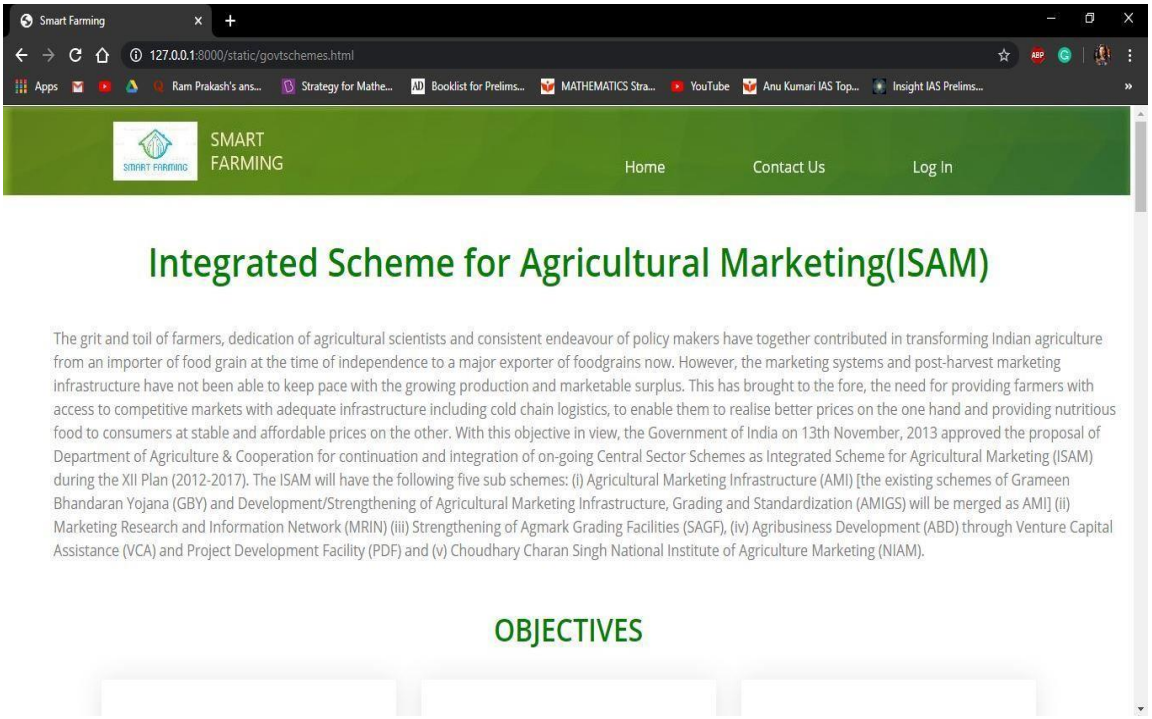


Figure 7.2 Site on Chrome browser

Internet Explorer:

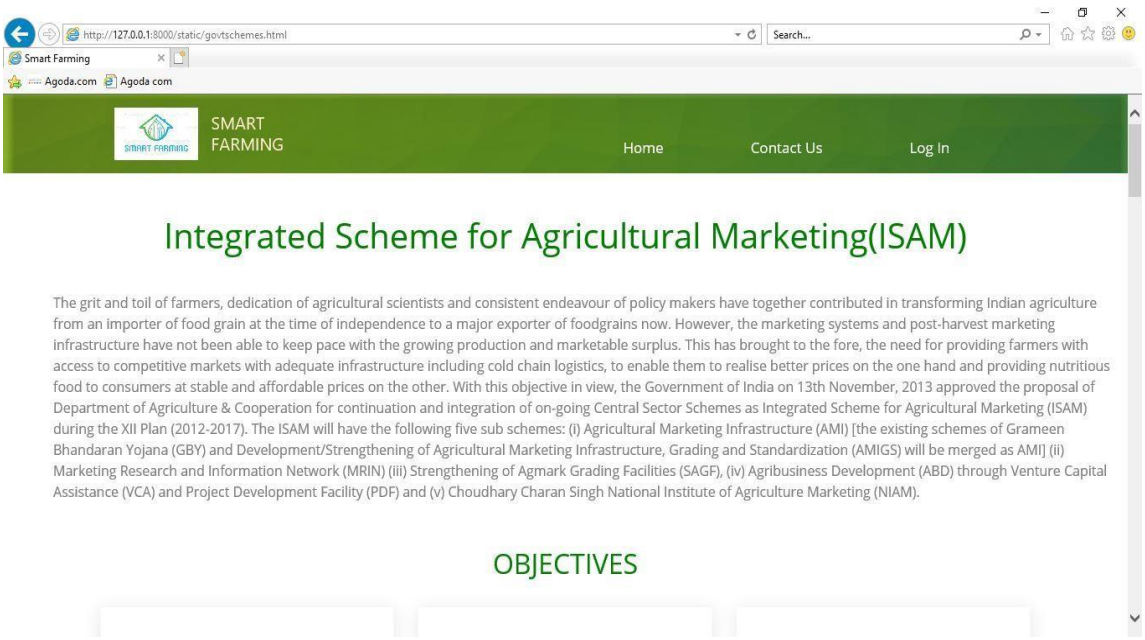


Figure 7.3 Site on Internet Explorer

7.2.3. The site is completely responsive and can be accessed using any mobile device and PCs.

All the components, menus, text, images are responsive and are adjusted according to the device's resolution. Following screenshots show that site can be opened through multiple devices having different screen resolutions.

7.2.3.1. Laptop:

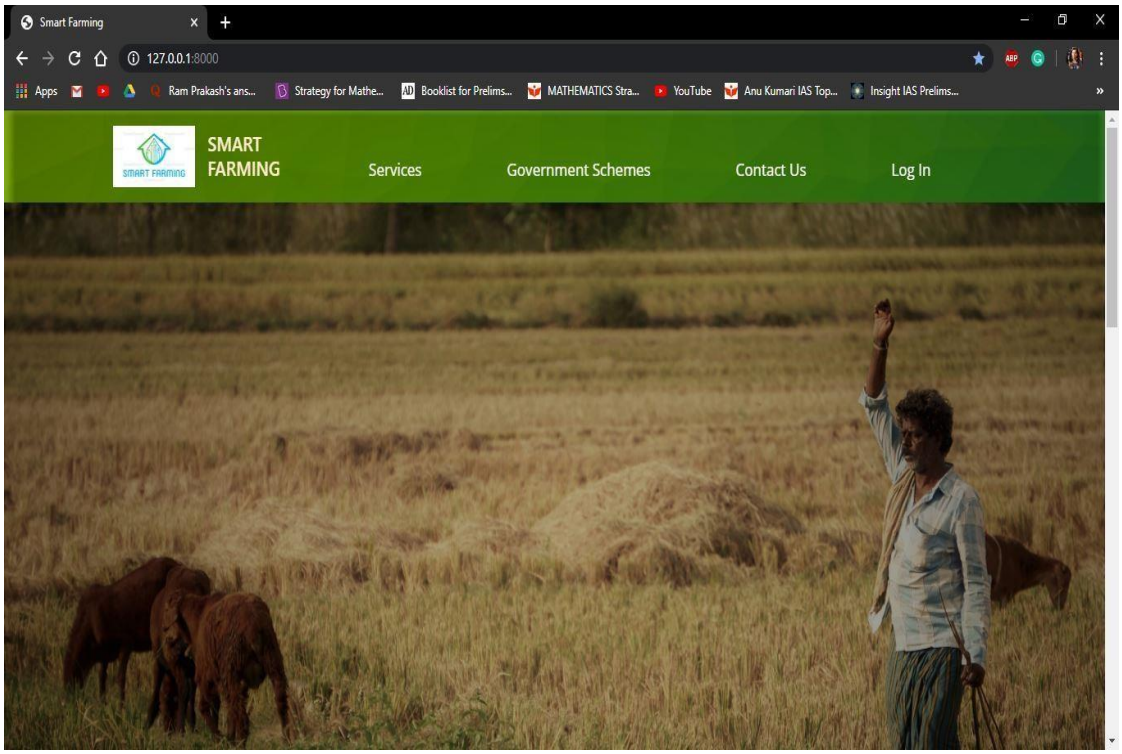


Figure 7.4 Site on laptop

7.2.3.2. Mobile (Resolution – 360 * 640):



Figure 7.5 Site on Mobile Phone

7.2.3.3. Ipad/ Tab (Resolution – 768 * 1024):

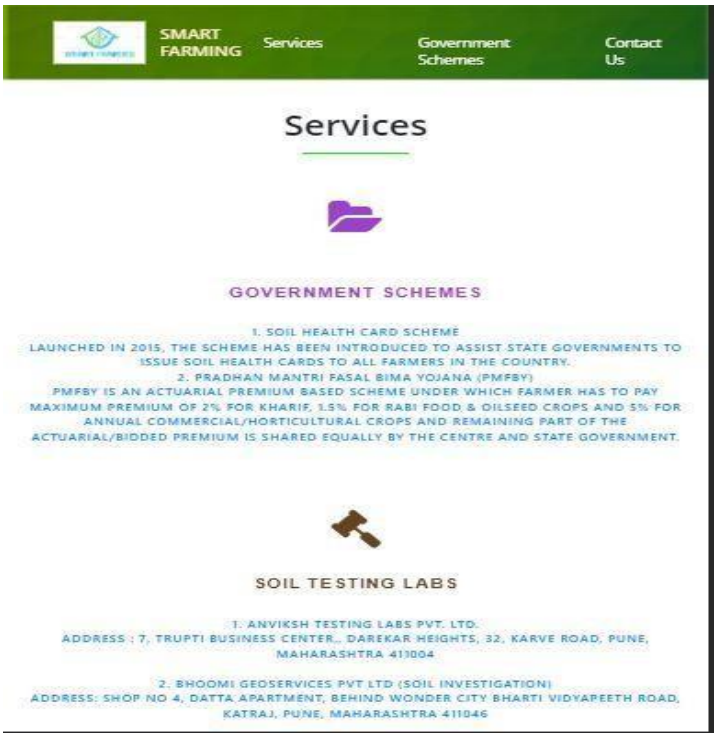


Figure 7.6 Site on Ipad/Tab

7.2.3.4. Mobile Menu Expand:

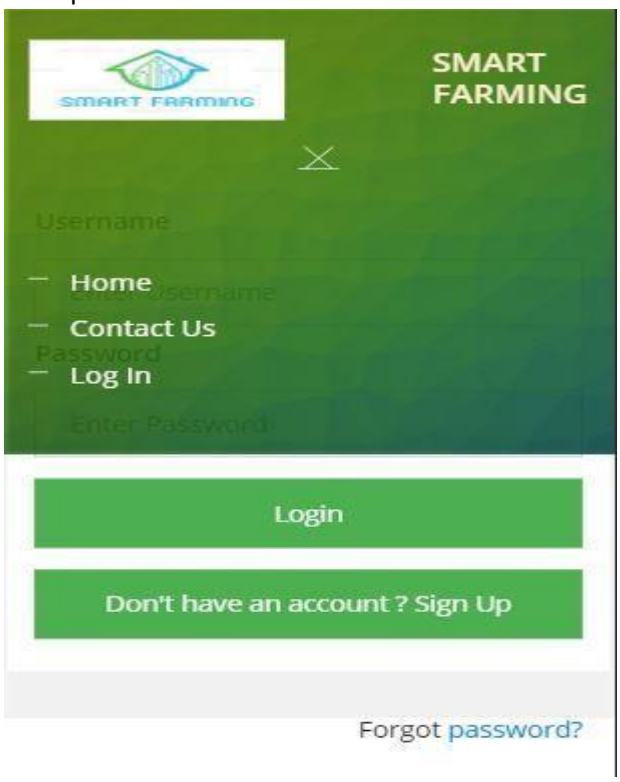
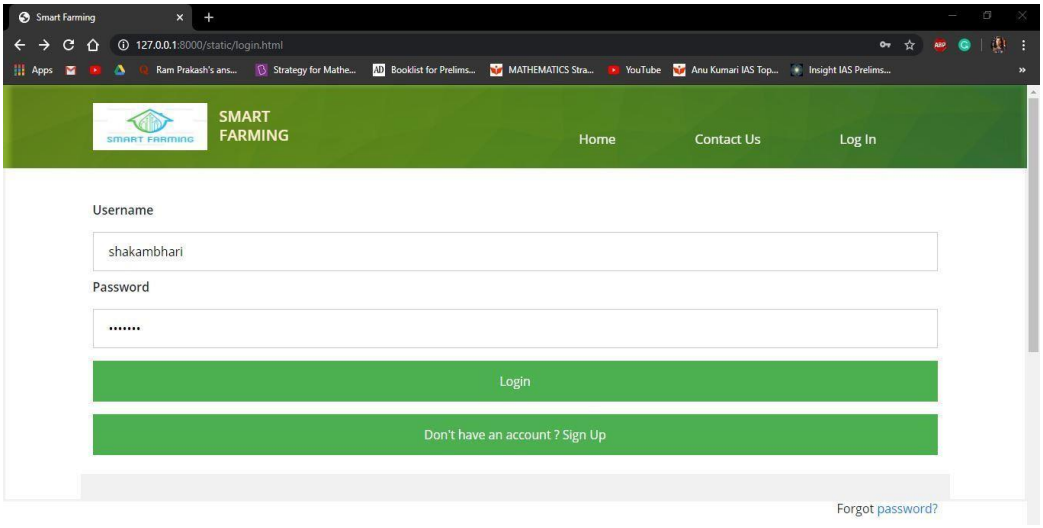
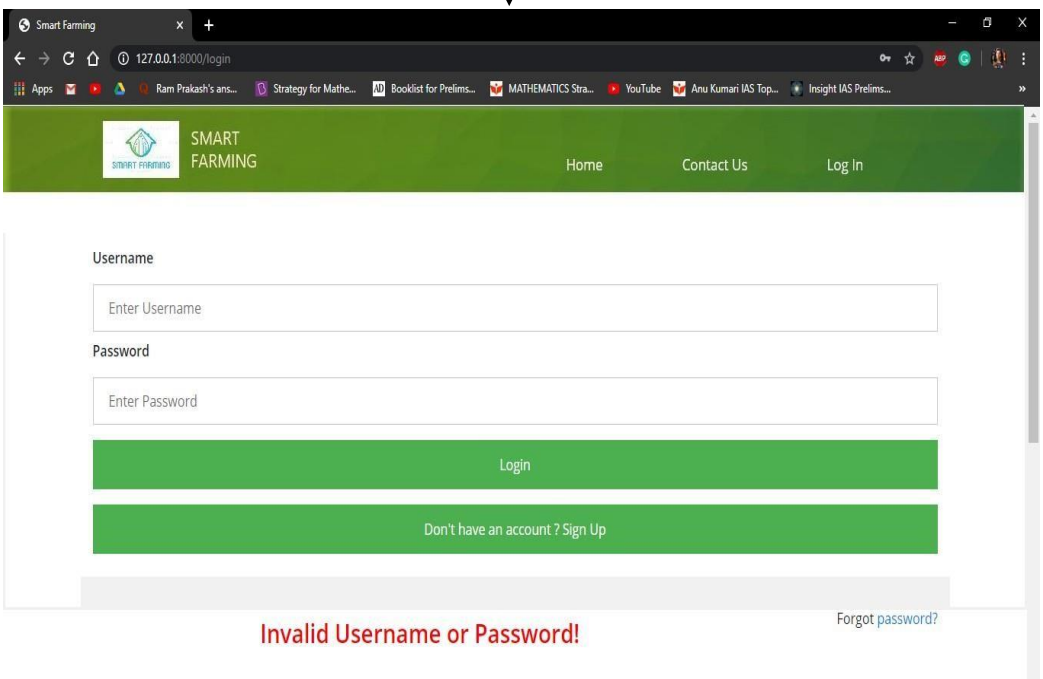


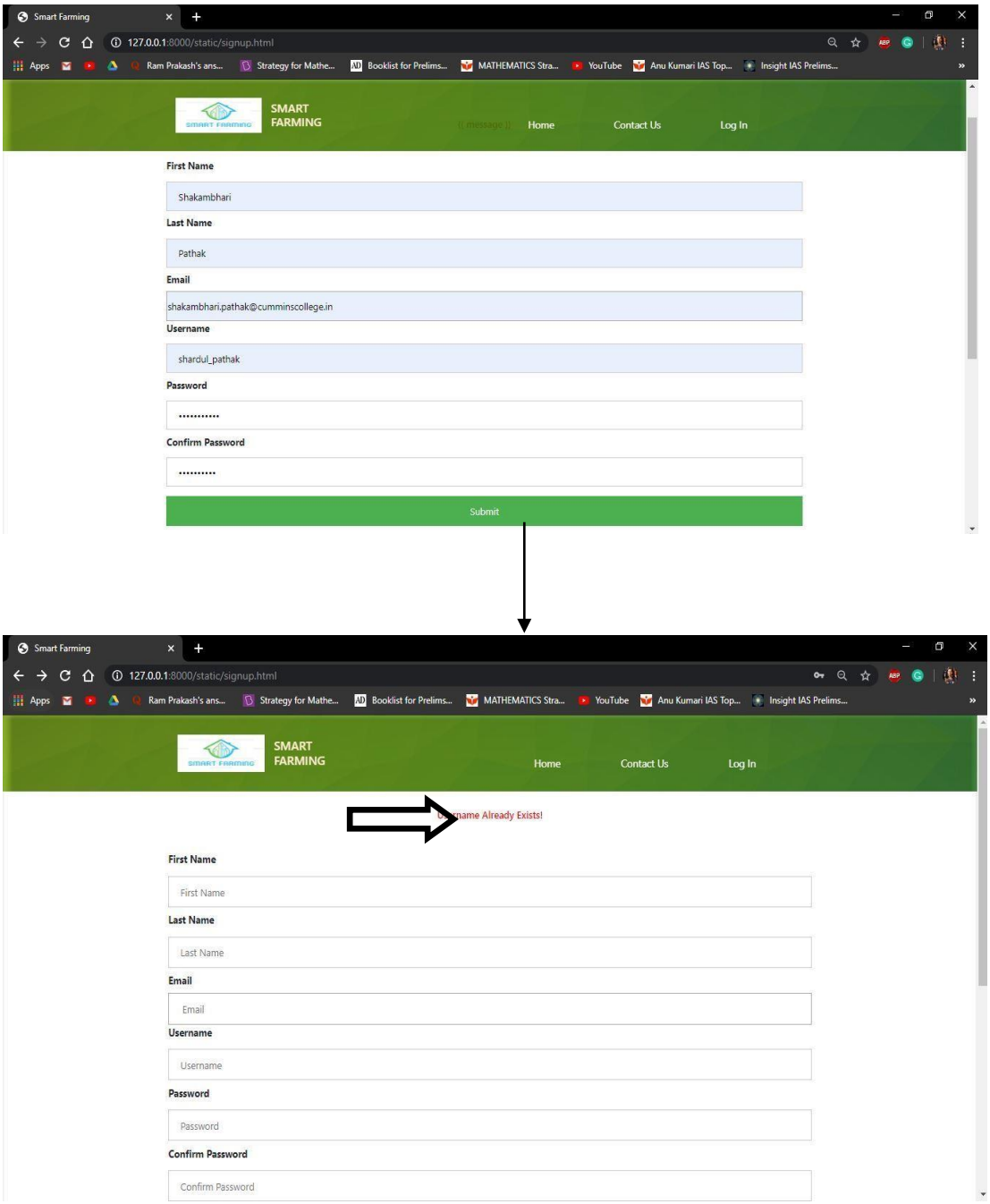
Figure 7.7 Menu Expansion

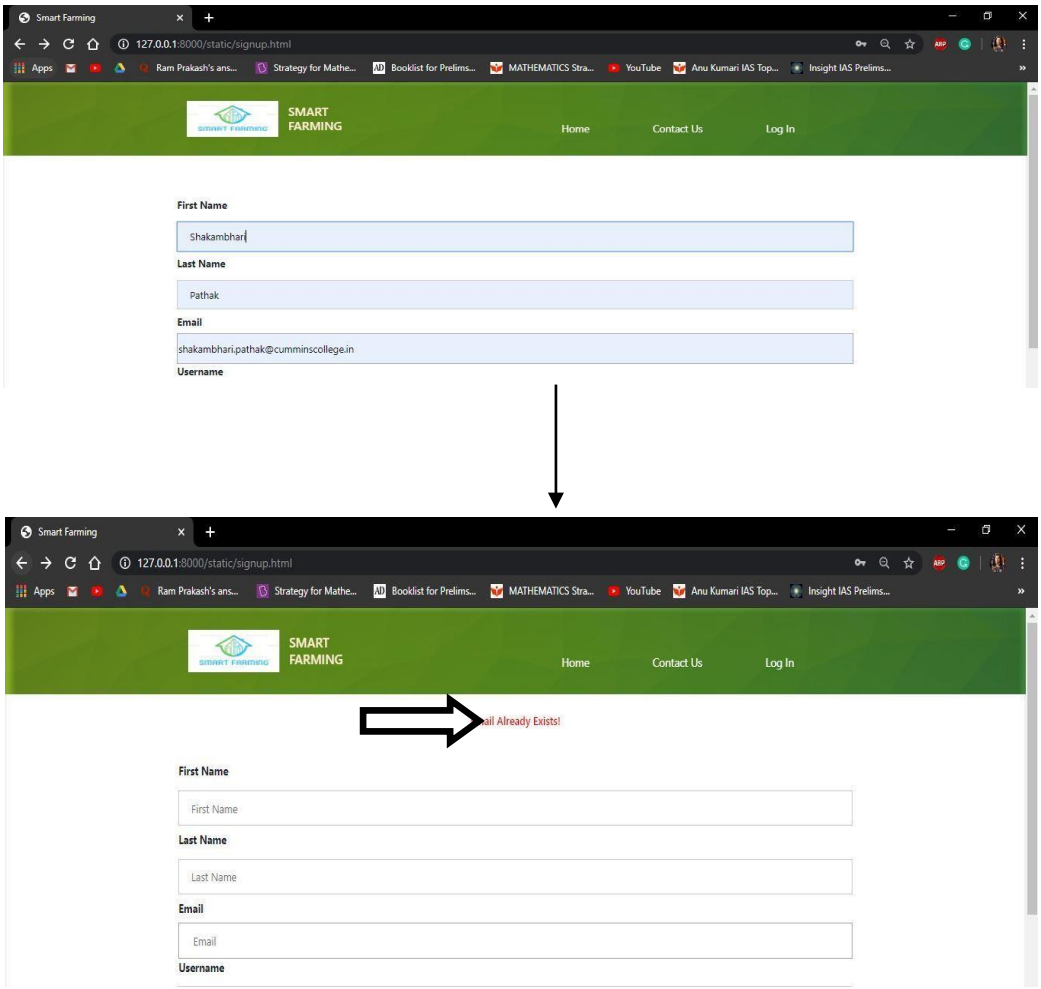
7.2.4. The user is able to login to the system only with valid credentials and the error message is shown if the user tries to login with wrong username or password. If the user enters a wrong username –



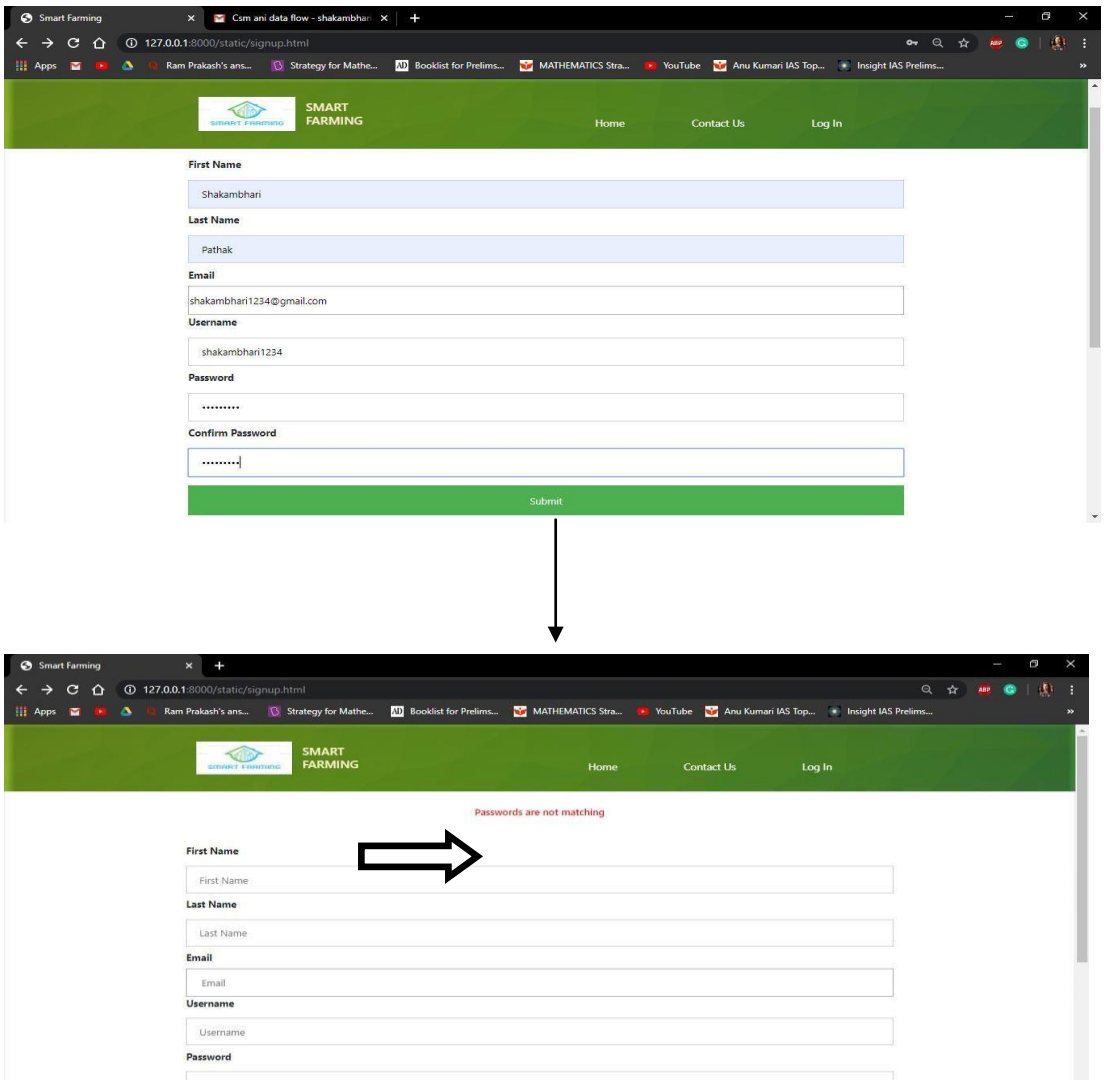


7.2.5. While signing in user won't be able to repeat the username or email that is already used. If any user tries to enter the same username or email, he/she will be given an error message that email or username is repeated.

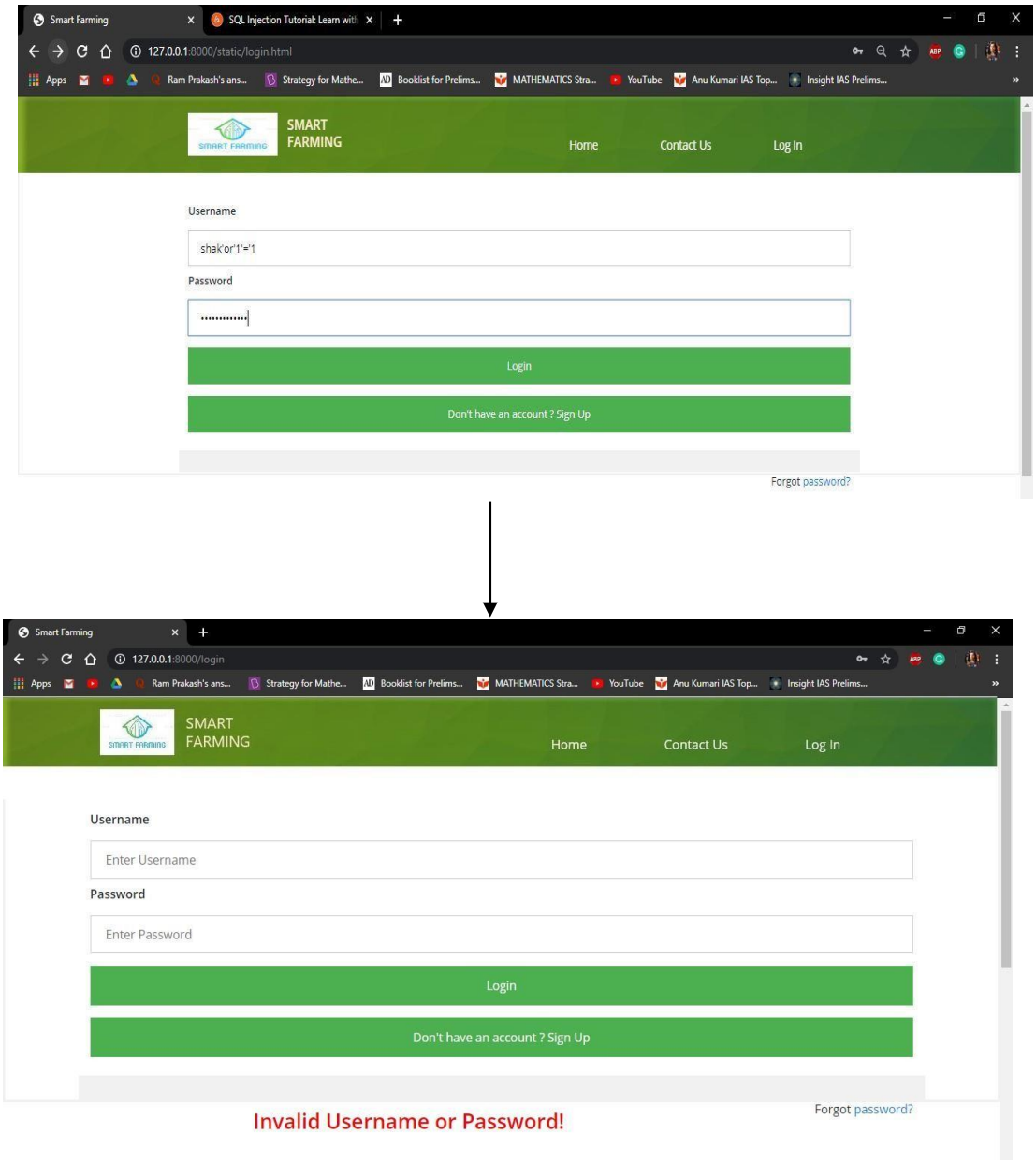




7.2.6. If user enters different passwords in ‘Password’ and ‘Confirm Password’ fields then he/she will be notified with an error message.



7.2.7. The site provides prevention from SQL Injection Attack.



7.2.8. Passwords are encrypted using hashing while getting stored in the database. Passwords are stored securely using hashing



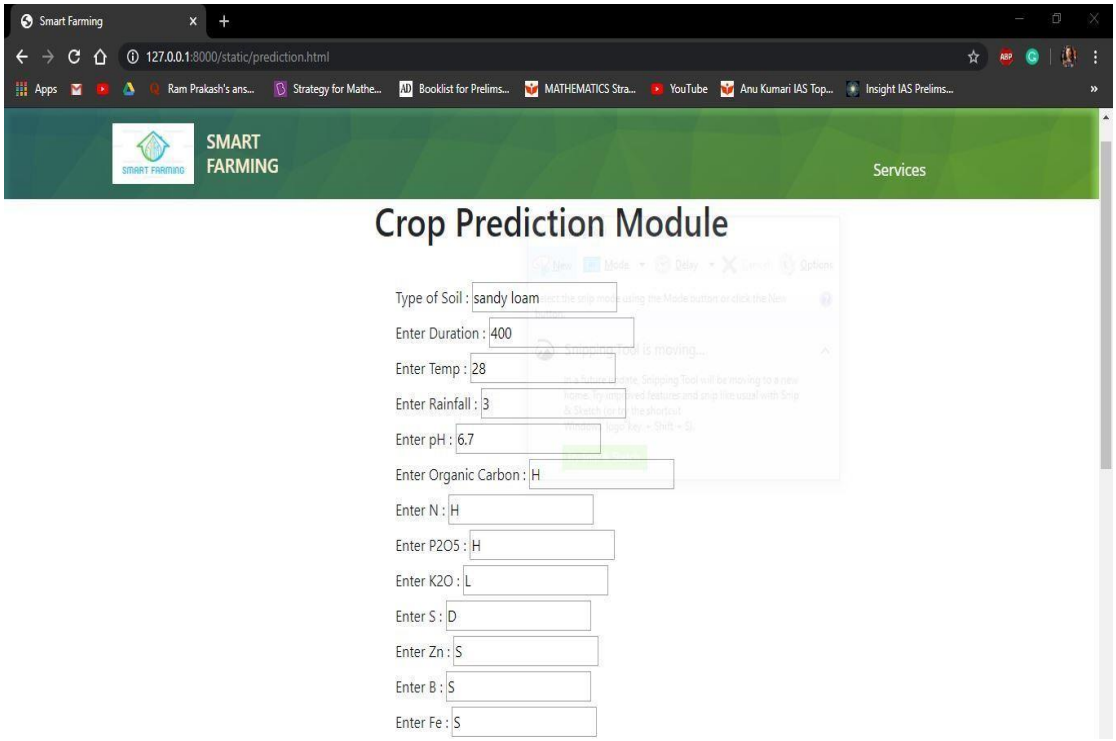
CHAPTER 8: RESULTS

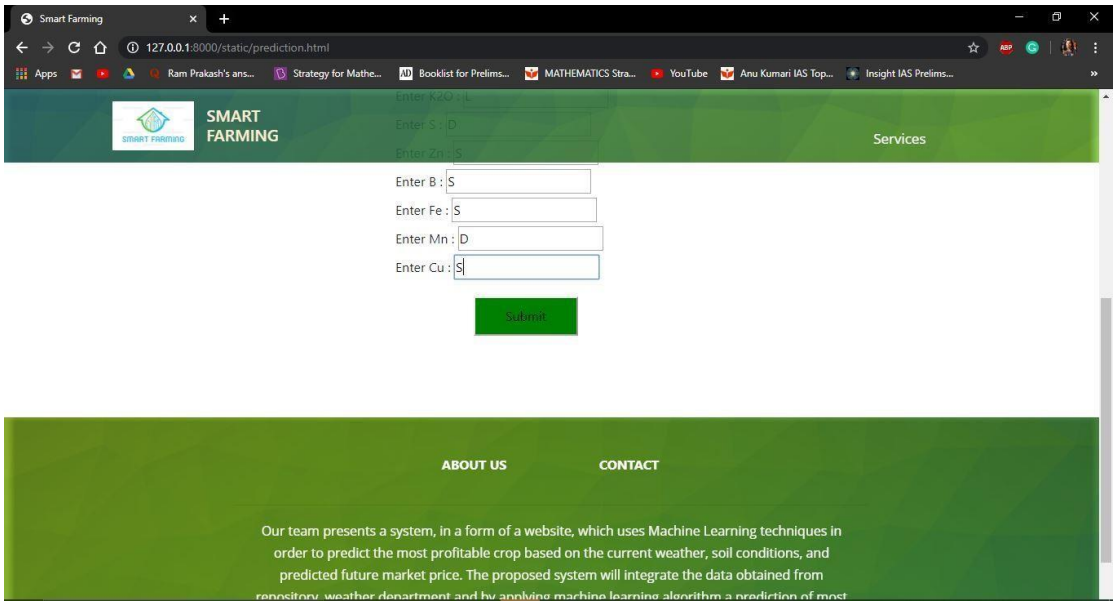
8.1 OUTCOMES AND SCREENSHOTS

8.1.1 Modules Results

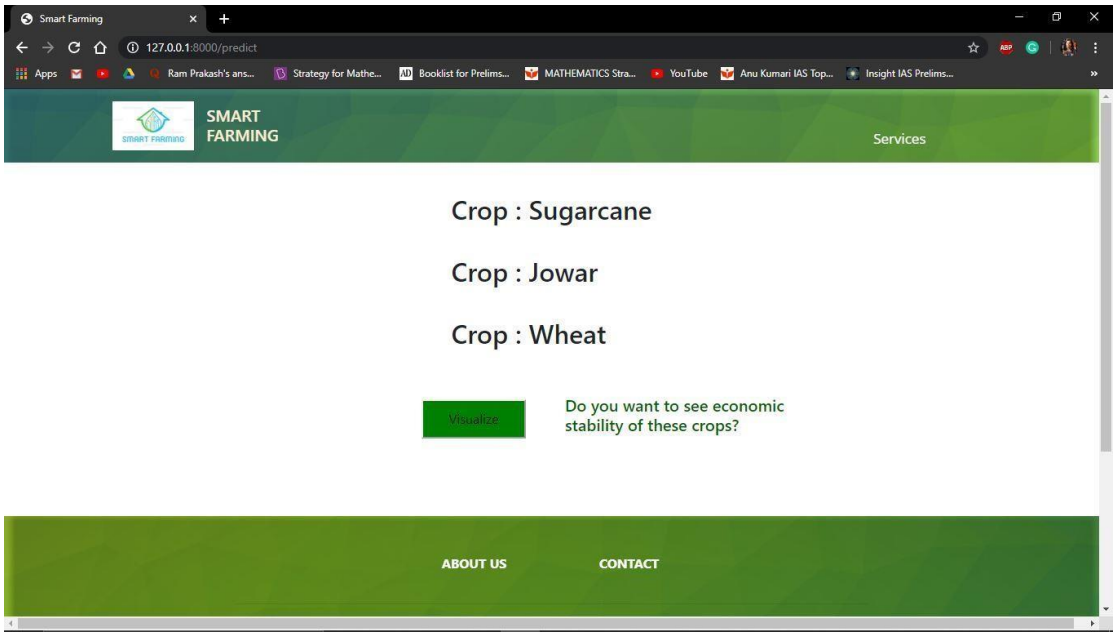
Module 1: Top 3 crop recommendations

The model recommends top 3 crops to the user based upon his land properties – namely the n, p, k, pH, temperature and other values along with the micronutrients.
Other factors like regional rainfall, soil type, etc are also considered.



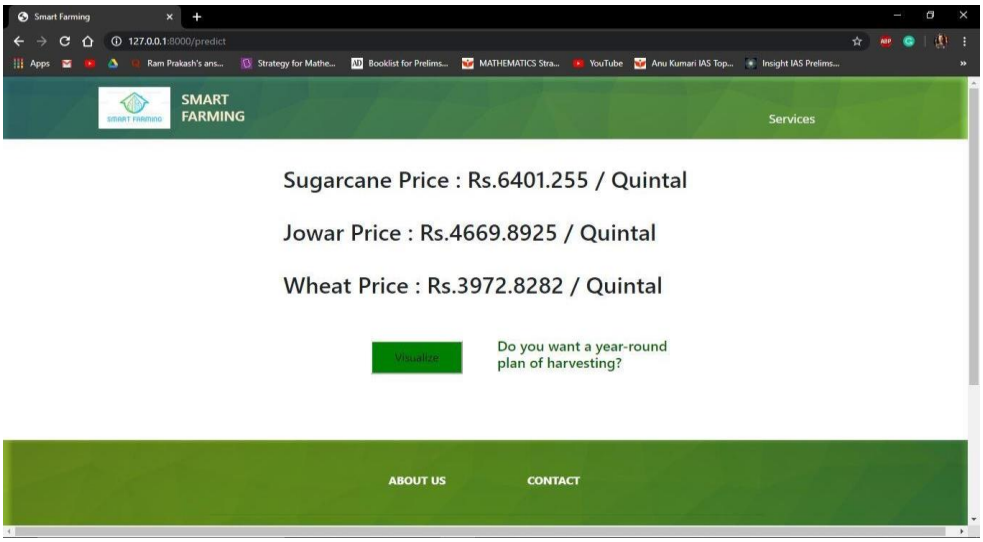


Result:



Module 2: Economically most beneficial crop (Prices prediction)

The prices of the top 3 crops recommended from above module are predicted. The one with the highest profit i.e economically most beneficial crop is finally recommended.



8.1.2 Algorithms Results Tool –

WEKA

Multiple Linear Regression

Random Tree.

Weka is one of the easiest, simple, widely used and tested open source machine learning exploration software. It contains provisions for data pre-processing, implementing classification algorithms, regression algorithms, clustering algorithms, understanding association rules, and provides visualization.

- Thus we have implemented multiple linear regression, a supervised machine learning algorithm. Here dataset containing fields relative to recommendation of crop based on land features was used. It included fields like NPK values, average rainfall value, temperature and the suitable crop (class label). Further crops were given certain

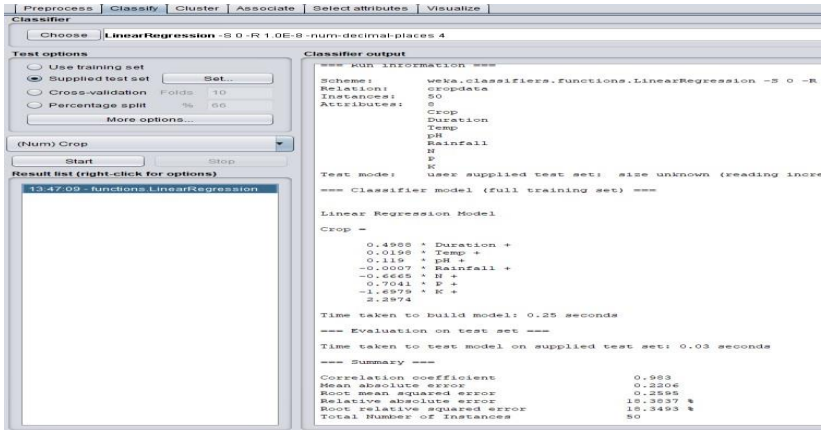


Figure 8.1 WEKA Tool

WEKA. It helped us understand the impact of each field on predicted output.

Thus in the above picture as we can see, say 0.119 units of rise in temp affects 1 unit prediction of crop. Also the mean absolute error, root mean squared error, etc of algorithm can be studied. This will help us compare various algorithms and its efficiency.

• The second algorithm implemented was random tree. The implementation procedure was similar to that of multiple linear regression. The extra functionalities explored specifically of decision tree with the help of weka tool was to easily identify the split decision that tool identified. This could be done by effective visualization provided by tree.

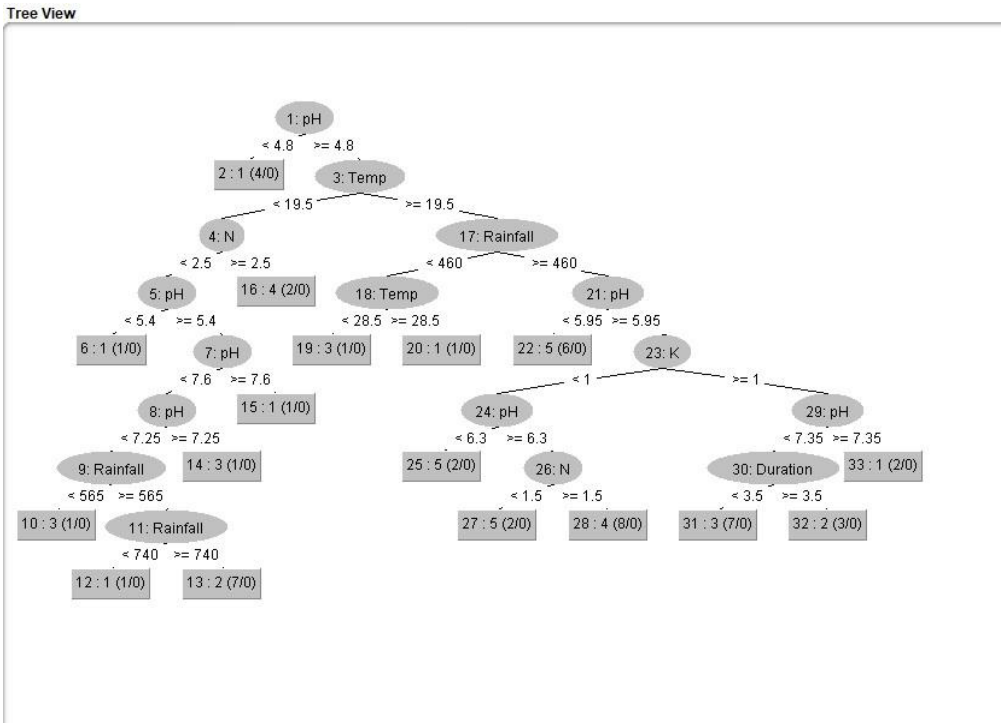


Figure 8.2 Decision Tree Algorithm

Thus the tree shows that the algorithm has used pH as the first split field. And following various decisions the leaf node will be the recommended crop. Here a supervised classification technique is used to classify the input tuple into its respective class of crop.

This module also displays the error values to be considered as metrics for decision of selection of algorithm.

Tool – Scikit library python

Multiple Linear Regression

Logistic Regression

Decision Tree

K Nearest Neighbours

Scikit-learn are a data science based library in Python that provides us with many unsupervised and supervised learning algorithms to explore and learn. It's built upon the technology stack like NumPy, pandas, and Matplotlib. The functionality that scikit-learn provides us include: Regression – including Linear regression and Logistic Regression, Classification – including K-Nearest Neighbors, Clustering – including Kmeans and K-Means++, Model selection, Preprocessing – including Min-Max Normalization.

The dataset imported for implementation of these algorithms contained fields relative to recommendation of crop based on land features. It included fields like NPK values, average rainfall value, temperature and the suitable crop (class label). Further crops were given certain respective numeric values say 1 for bajra, 2 for rice, etc. Data was divided into training and testing tuples. And it was preprocessed to scale the values. This was common in all the algorithms. Further individual algorithms were implemented by using in built methods of the scikit library. Accuracy output of individual algorithms was observed to be:

• Multiple Linear Regression



Figure 8.3 Multiple Linear Regression

Observed error and the variance between actual and predicted values is thus seen. A visual graph is generated using matplotlib.

- Decision Tree

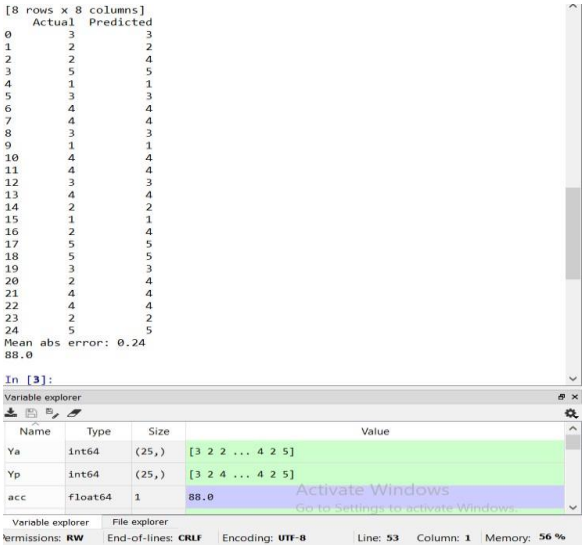


Figure 8.4 Decision Tree

- K Nearest Neighbours

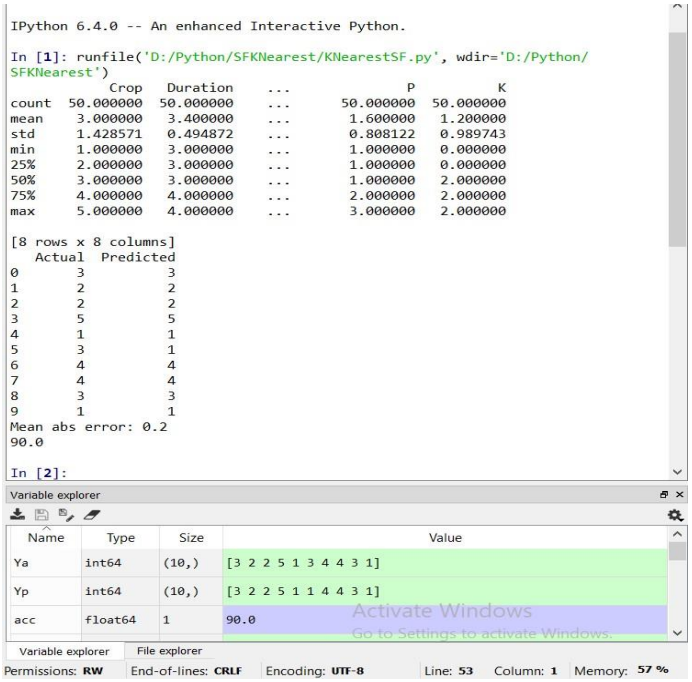


Figure 8.5 KNN

Comment:

APIs of the above mentioned algorithms were generated. Thus instead of loading the entire dataset and running the code on client side, the api is generated on server side beforehand. Further only the api is called on the client side.

Logistic regression was the choice of algorithm for the base model due to its better prediction and higher rate of accuracy.

Other algorithms are kept open for researcher for study and exploration.

CHAPTER 9: CONCLUSION

9.1 CONCLUSION

Summing up, as the last year project an attempt is made to bridge the gap between labour, finance, land etc. invested in agriculture and yet its lower contribution to GDP. The datasets needed for the project have been gathered. It includes datasets with fields of land properties, environmental factors as well as the future price consideration of crops. Interactions with various agricultural colleges, labs and experts have been done for the same. The functioning of entire system is designed in a way to include:

- Economically beneficial Crop recommendation
- Crop pattern decision support through a particular time
- Conditions and precautions to be observed for a particular crop and particular ownership
- Awareness on various government schemes
- Other simple but influential functionalities

As mentioned, data analytics and machine learning are the main domain areas focused on technically. Learning exposure in fields of deep learning and UI development through usage of Django is targeted. Whereas learnings done till now include:

- On field data acquisition
- Understanding of Soil Health Card
- Exploring pre-processing tool (rapidminer).
- Algorithms implementation – scikit and tool (weka).
- Exploration of LUCID platform.
- Responsive web pages concept.

Interactions with MSC research students and Agriculture college officials to understand the problems in depth.

The algorithms to be worked upon have been decided and will help in establishing a new dimension in the proposed work. Gap identification and thus bridging it is what makes this an innovative approach to a real-world problem.

9.2 FUTURE WORK

Nothing is ever completely done. So a lot of aspects have been considered which reflect in the mentioned future scope:

- To include use of IOT sensors to automatically collect soil data.
- To increase granularity of rainfall prediction.
- To suggest suitable fertilizers and pesticides for the crop.
- To analyze and predict requirements of import and export.
- To consider market demand in prediction.
- To personalize application to a level where individual farmer can use it for private land.
- To suggest variance in crop production if adjacent lands decide upon the same crops.

9.3 APPLICATIONS

Smart Farming can have further applications like:

- Advertisement of soil testing laboratories through the web app which may result in some extra revenue and business value.
- Along with the base crops included in the model, similarly the app can be used for extended farming areas like horticulture, mushroom projects, fruits, other cash crops, etc
- Soil Health Card summarization can also be an extended application and functionality of the web app.

APPENDIX B



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UGC-CARE(Group-II)Approved & SCOPUS ACTIVE Journal

ACCEPTANCE LETTER TO AUTHOR

Dear Author,

With reference to your paper submitted **"Smart Farming using Machine Learning and Data Analytics"** we are pleased to accept the same for publication in JUSST - Journal of University of Shanghai for Science and Technology, **Volume 23, Issue 6, June-2021.**

Paper ID: JUSST/21/05-244

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- In case of any query please do not hesitate to contact us editor@jusst.org early reply is appreciated.
- Fee paid for the publication of the paper does not refund under any circumstances.

Date
2/6/2021

Best regards,
Dr. XuhuiFeng
jusst.org

Dr. XuhuiFeng
Editor-in-Chief

Member of Crossref

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